

BFS Traversal

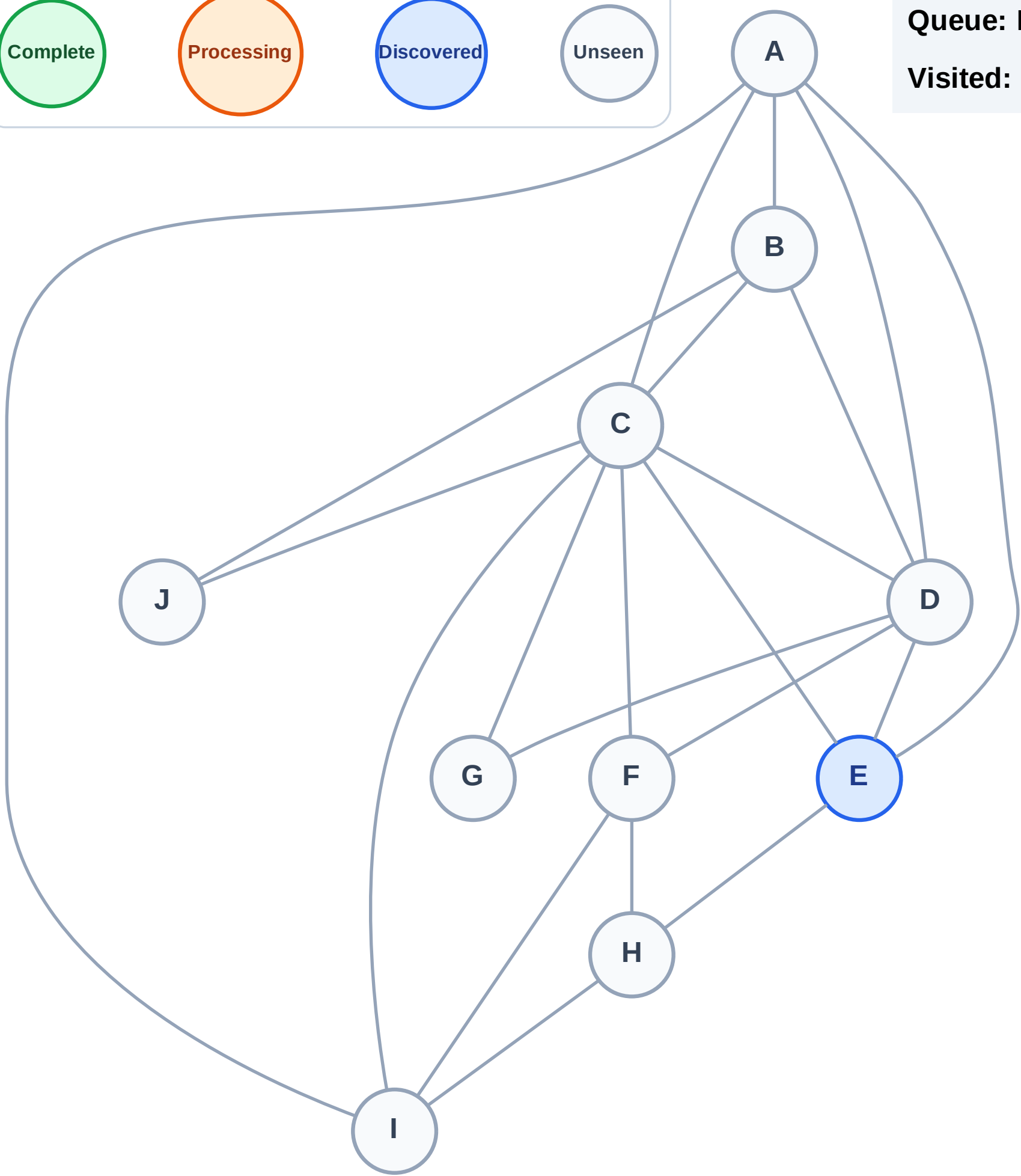
Step 1: Start at E

Legend



Queue: E

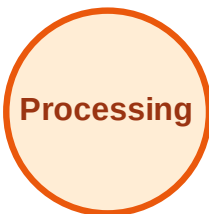
Visited: (none)



BFS Traversal

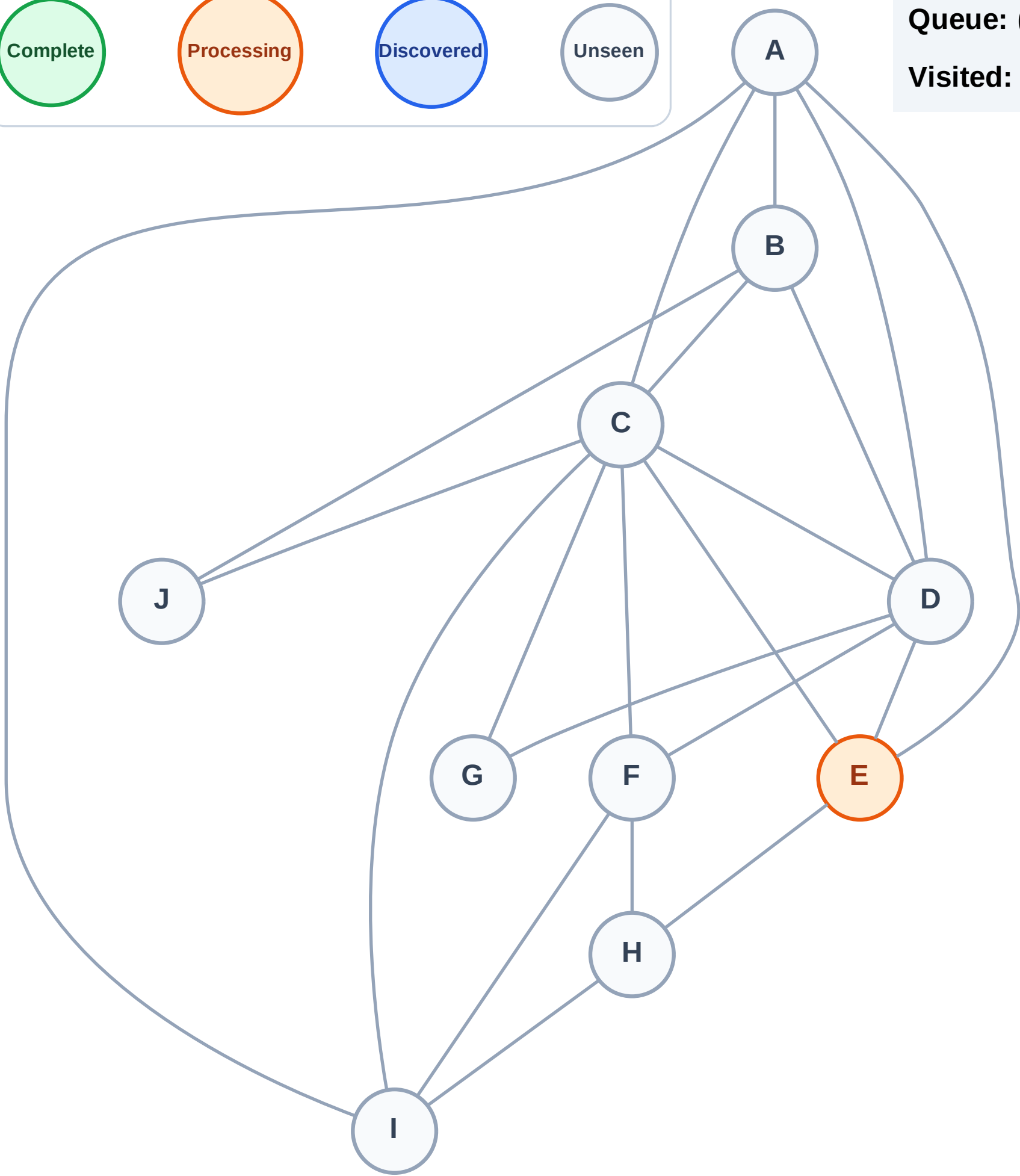
Step 2: Process node E

Legend



Queue: (empty)

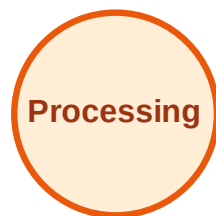
Visited: (none)



BFS Traversal

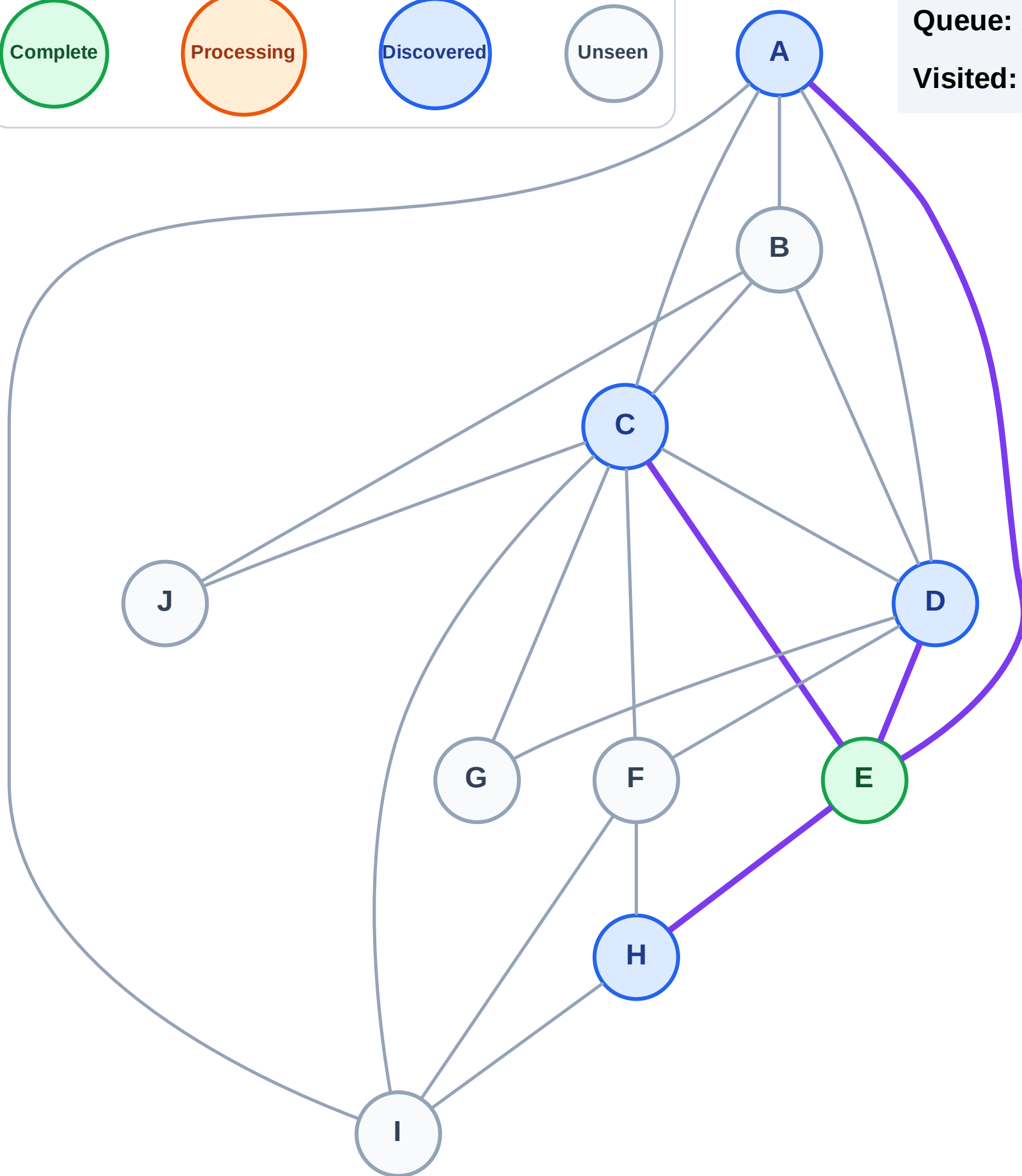
Step 3: Discover A, C, D, H from E

Legend



Queue: A → C → D → H

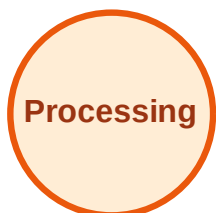
Visited: E



BFS Traversal

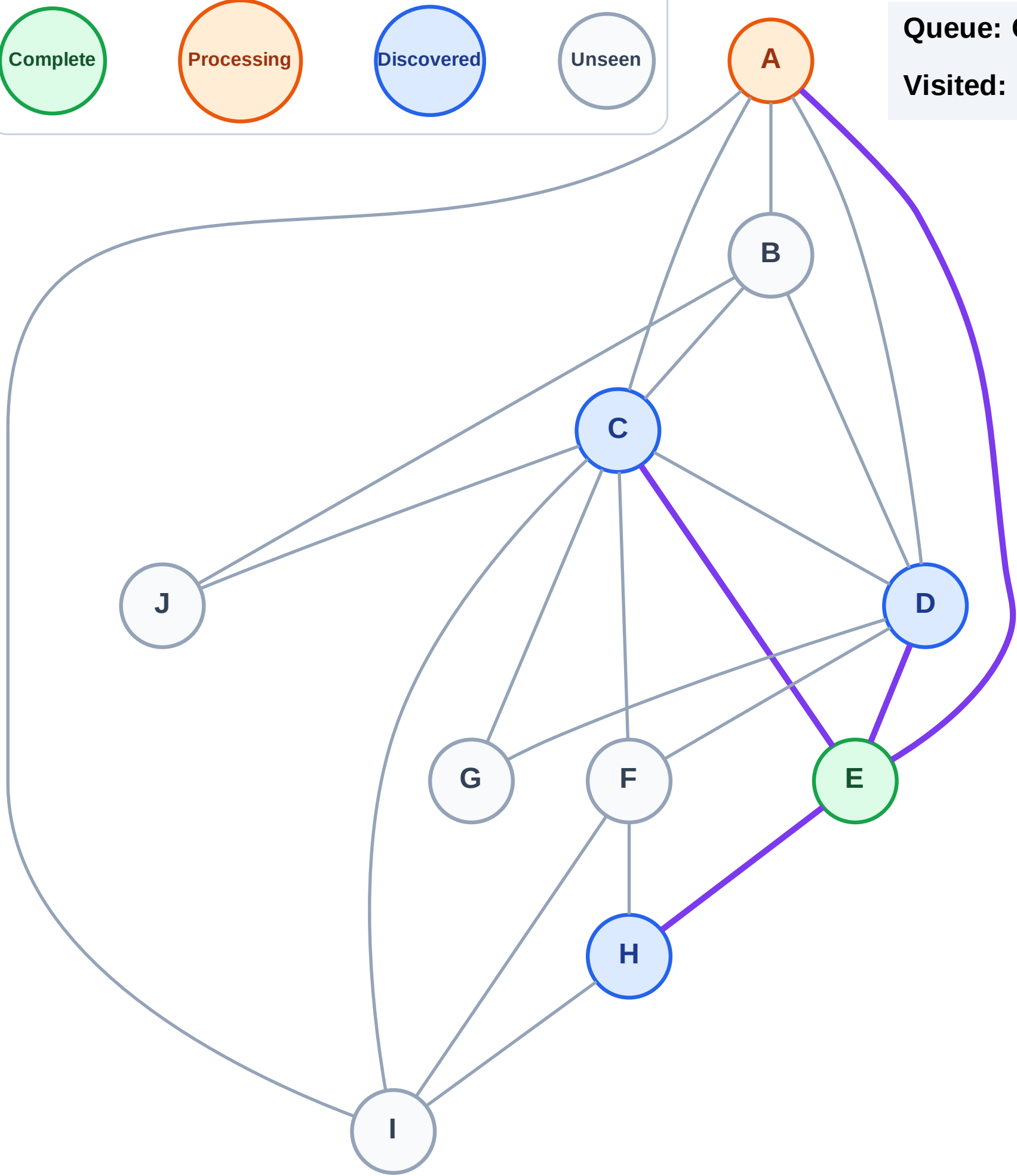
Step 4: Process node A

Legend



Queue: C → D → H

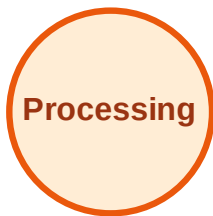
Visited: E



BFS Traversal

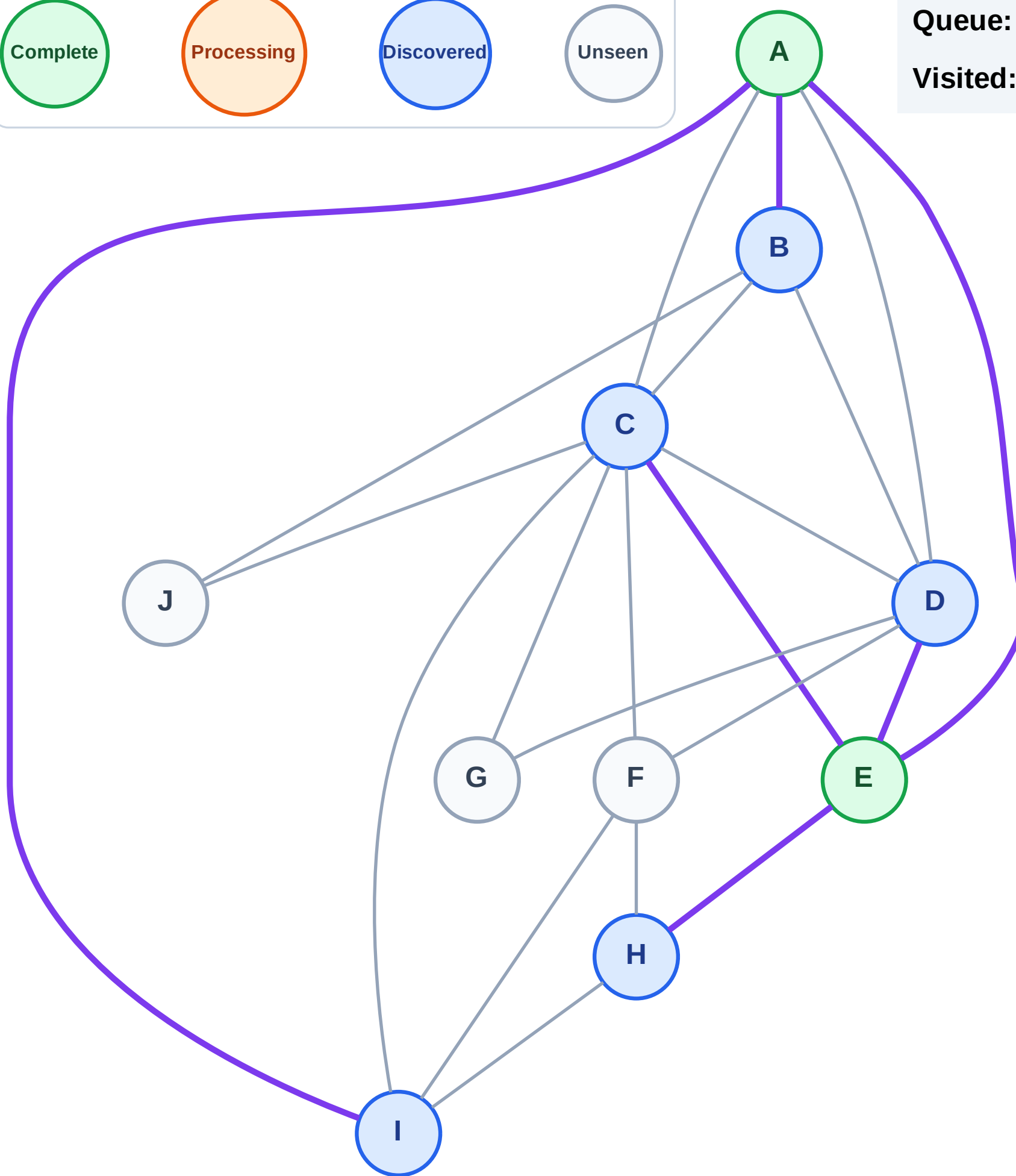
Step 5: Discover B, I from A

Legend



Queue: C → D → H → B → I

Visited: A, E



BFS Traversal

Step 6: Process node C

Legend

Complete

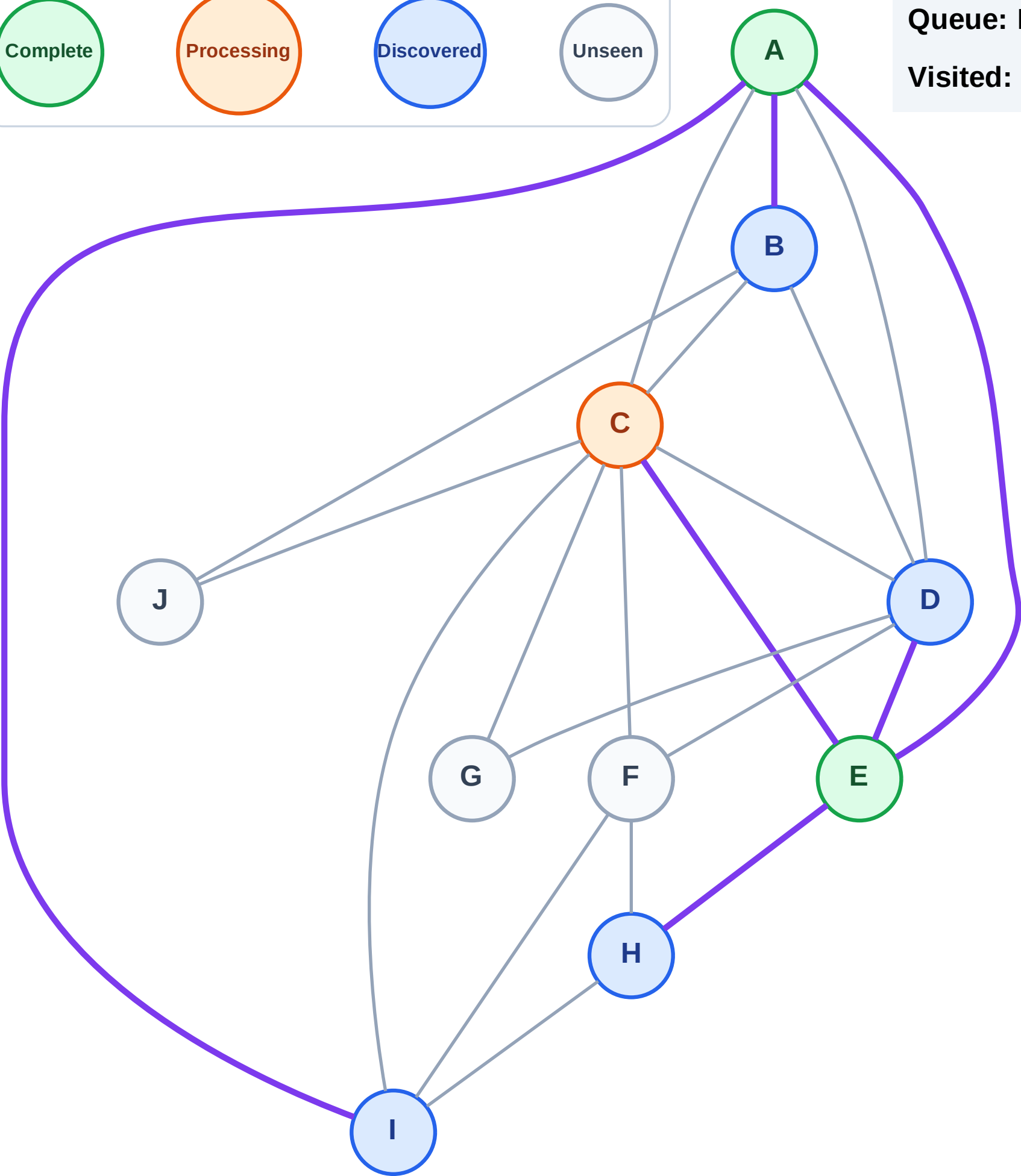
Processing

Discovered

Unseen

Queue: D → H → B → I

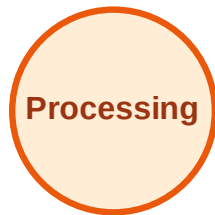
Visited: A, E



BFS Traversal

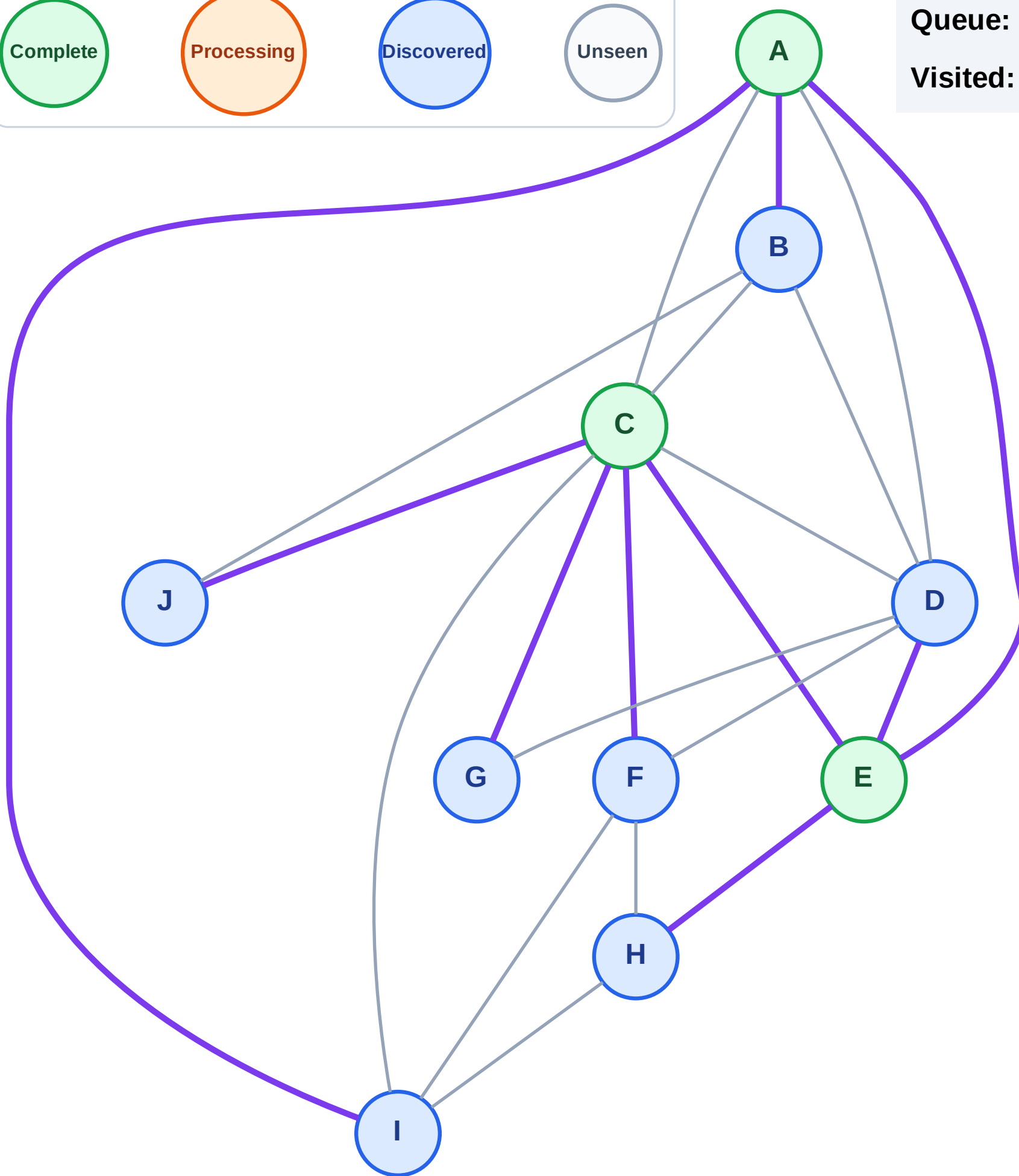
Step 7: Discover F, G, J from C

Legend



Queue: D → H → B → I → F → G → J

Visited: A, C, E



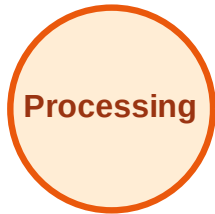
BFS Traversal

Step 8: Process node D

Legend



Complete



Processing



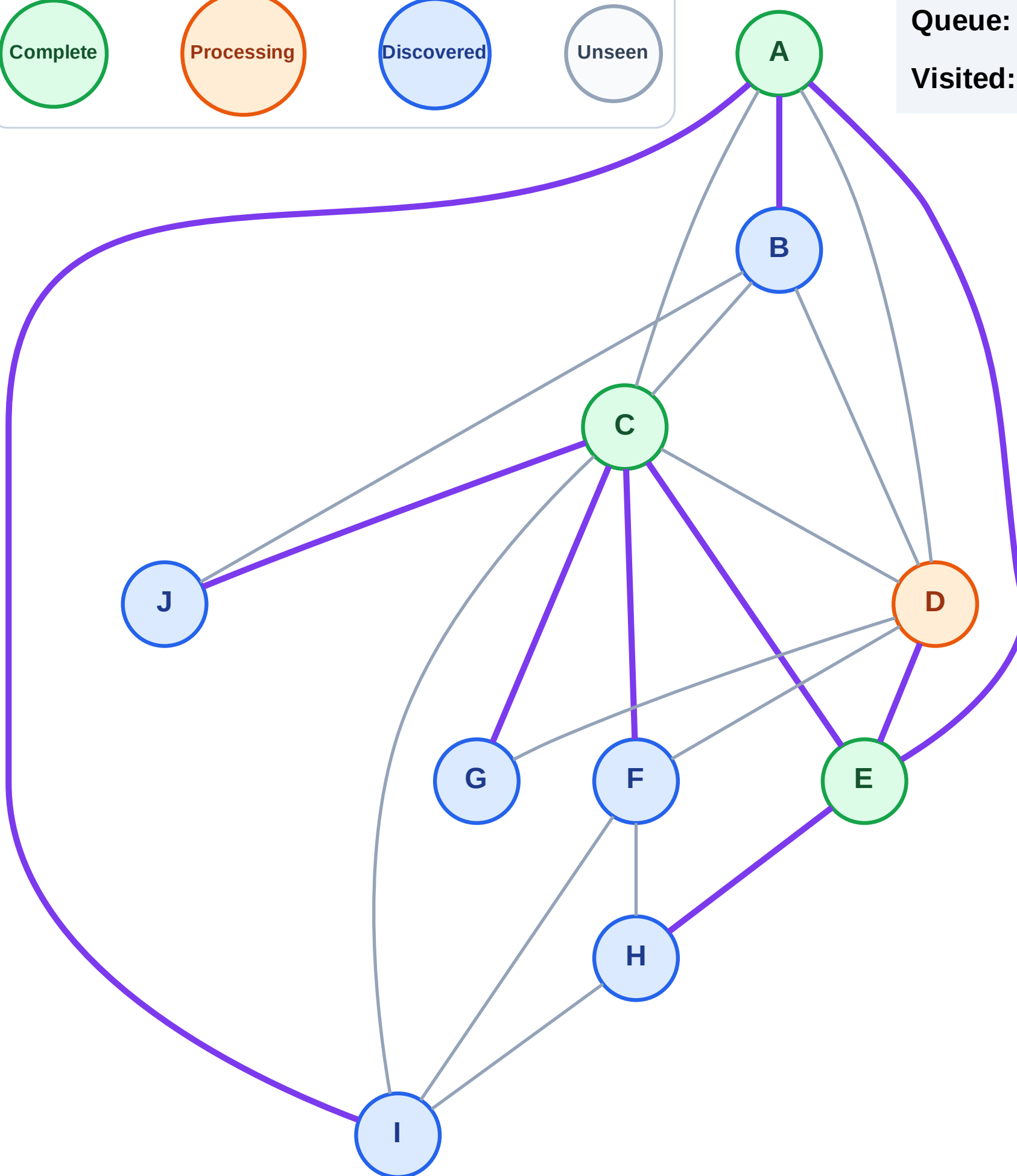
Discovered



Unseen

Queue: H → B → I → F → G → J

Visited: A, C, E



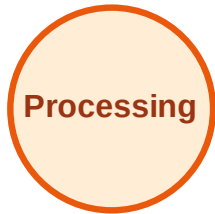
BFS Traversal

Step 9: No new neighbors from D

Legend



Complete



Processing



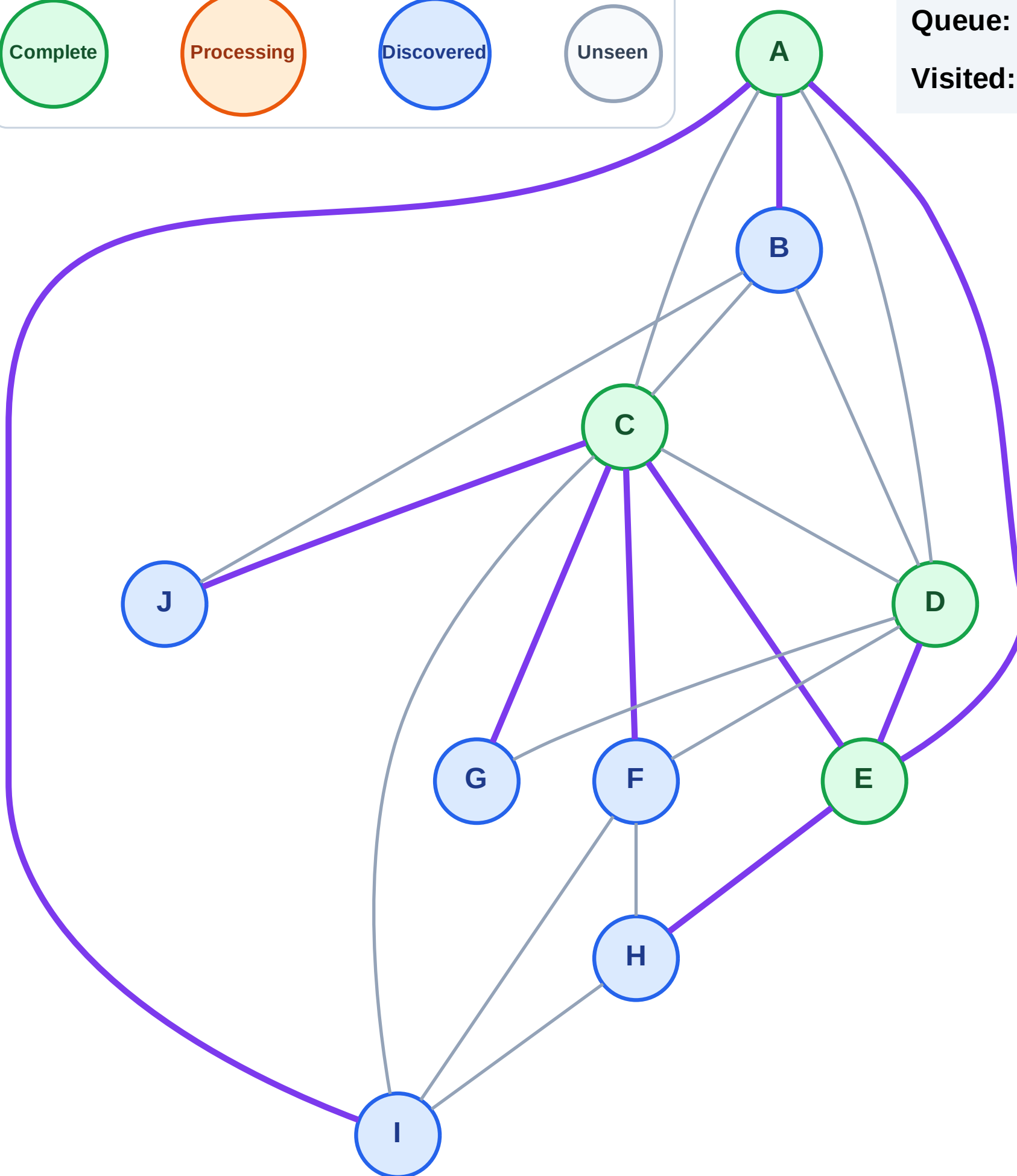
Discovered



Unseen

Queue: H → B → I → F → G → J

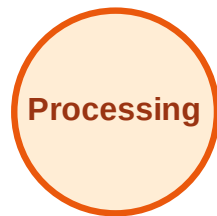
Visited: A, C, D, E



BFS Traversal

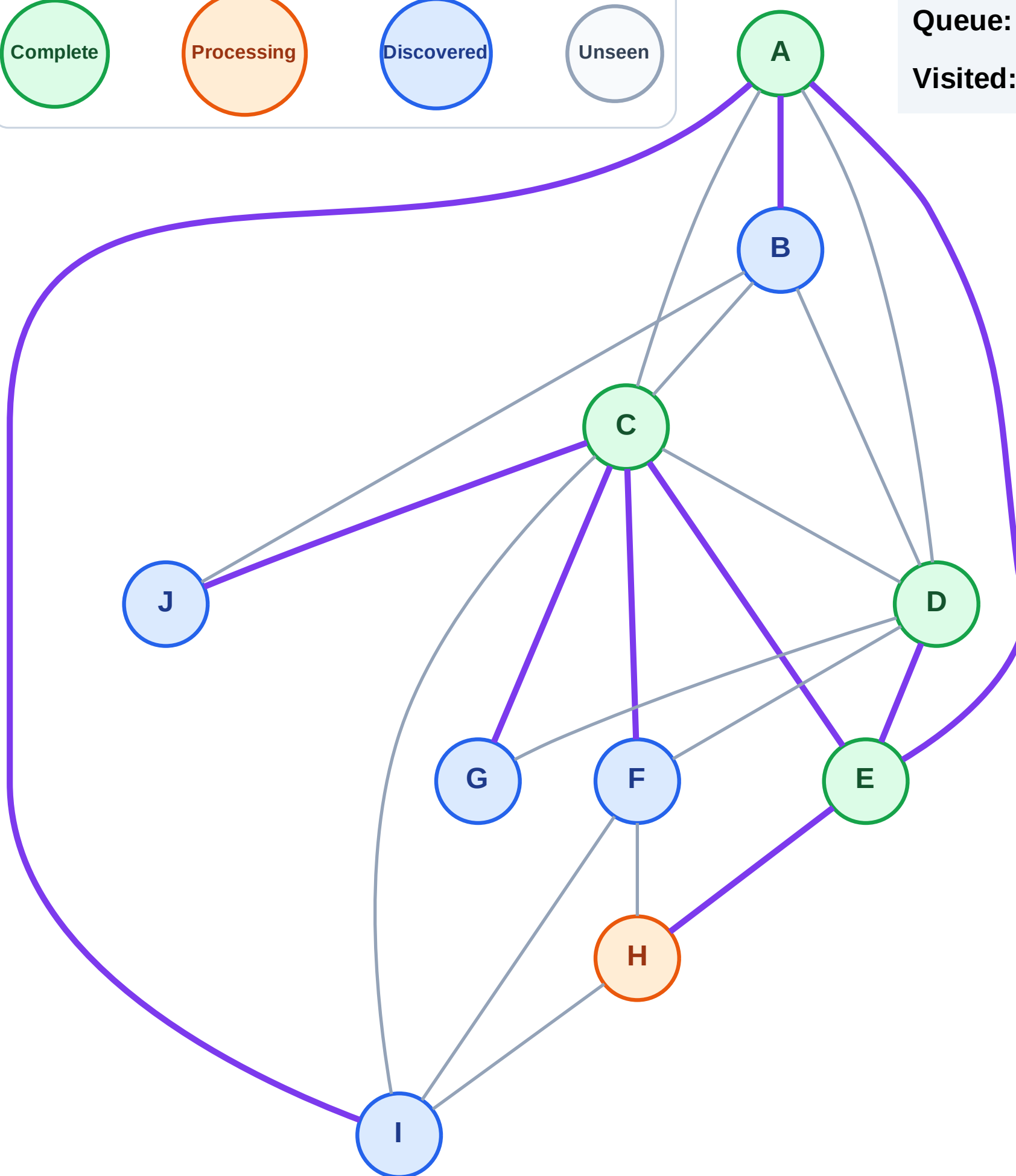
Step 10: Process node H

Legend



Queue: B → I → F → G → J

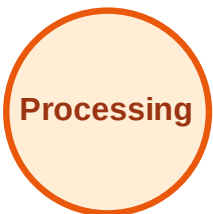
Visited: A, C, D, E



BFS Traversal

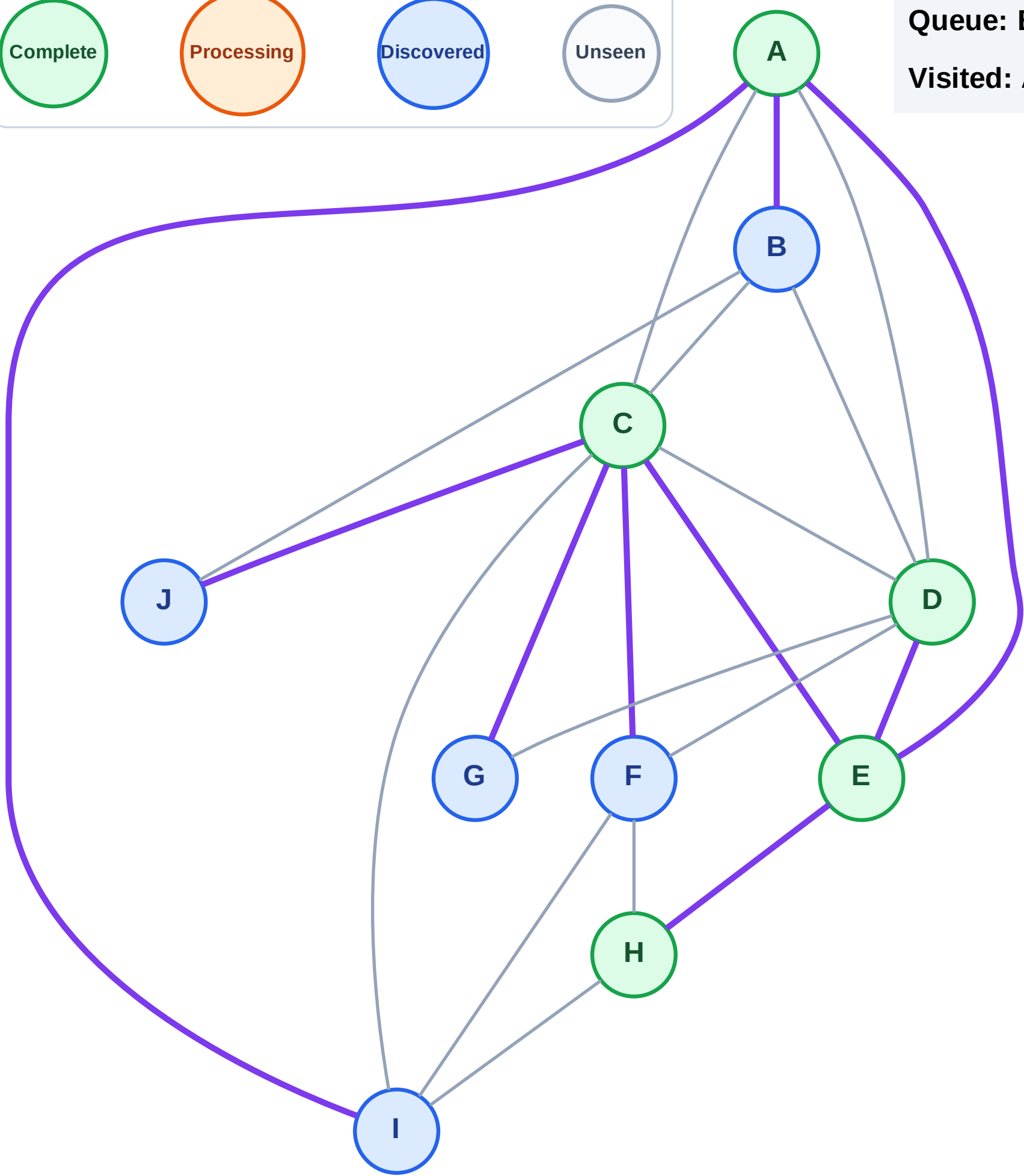
Step 11: No new neighbors from H

Legend



Queue: B → I → F → G → J

Visited: A, C, D, E, H



BFS Traversal

Step 12: Process node B

Legend

Complete

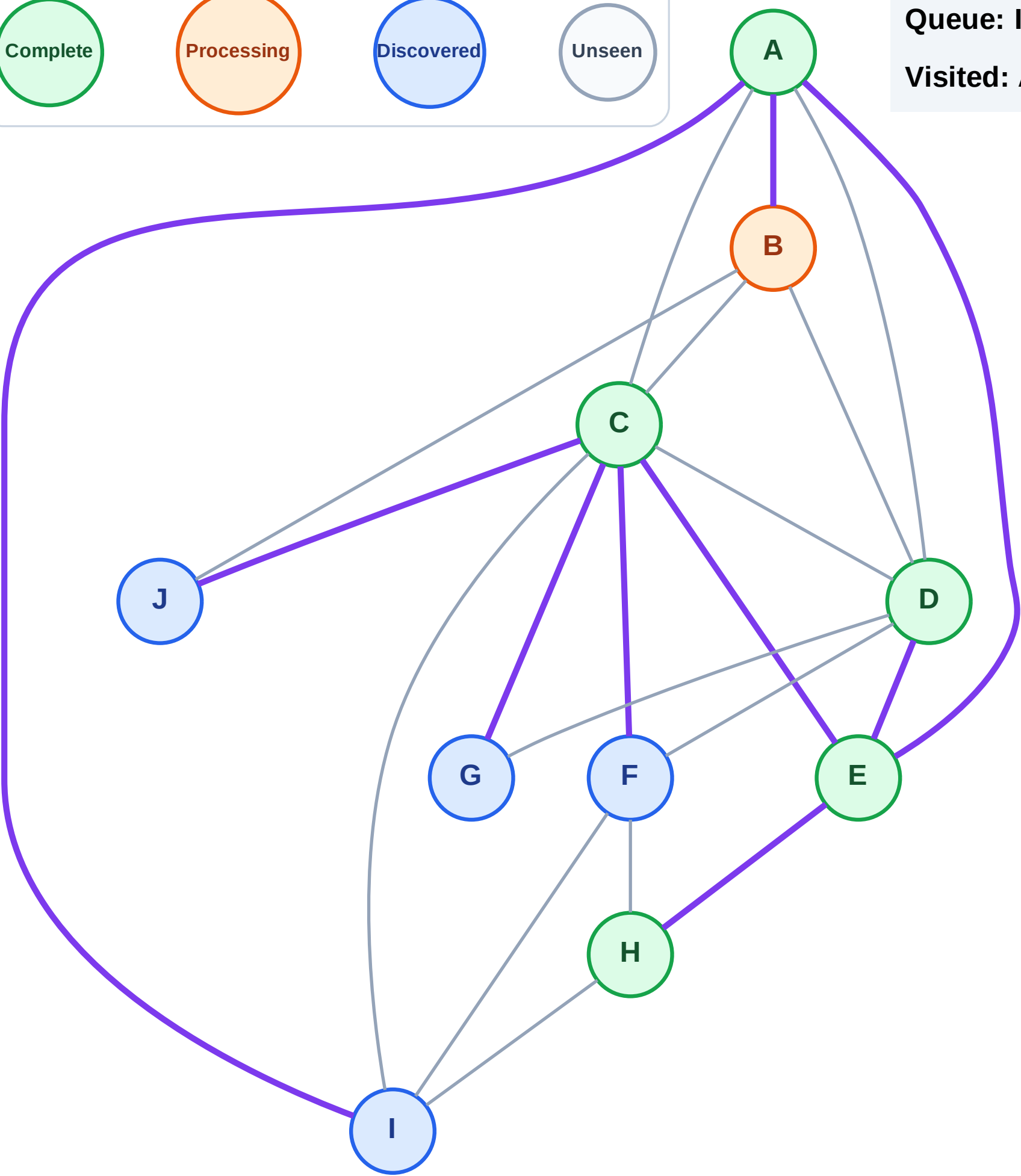
Processing

Discovered

Unseen

Queue: I → F → G → J

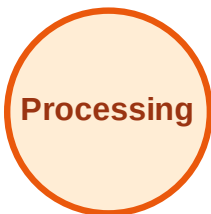
Visited: A, C, D, E, H



BFS Traversal

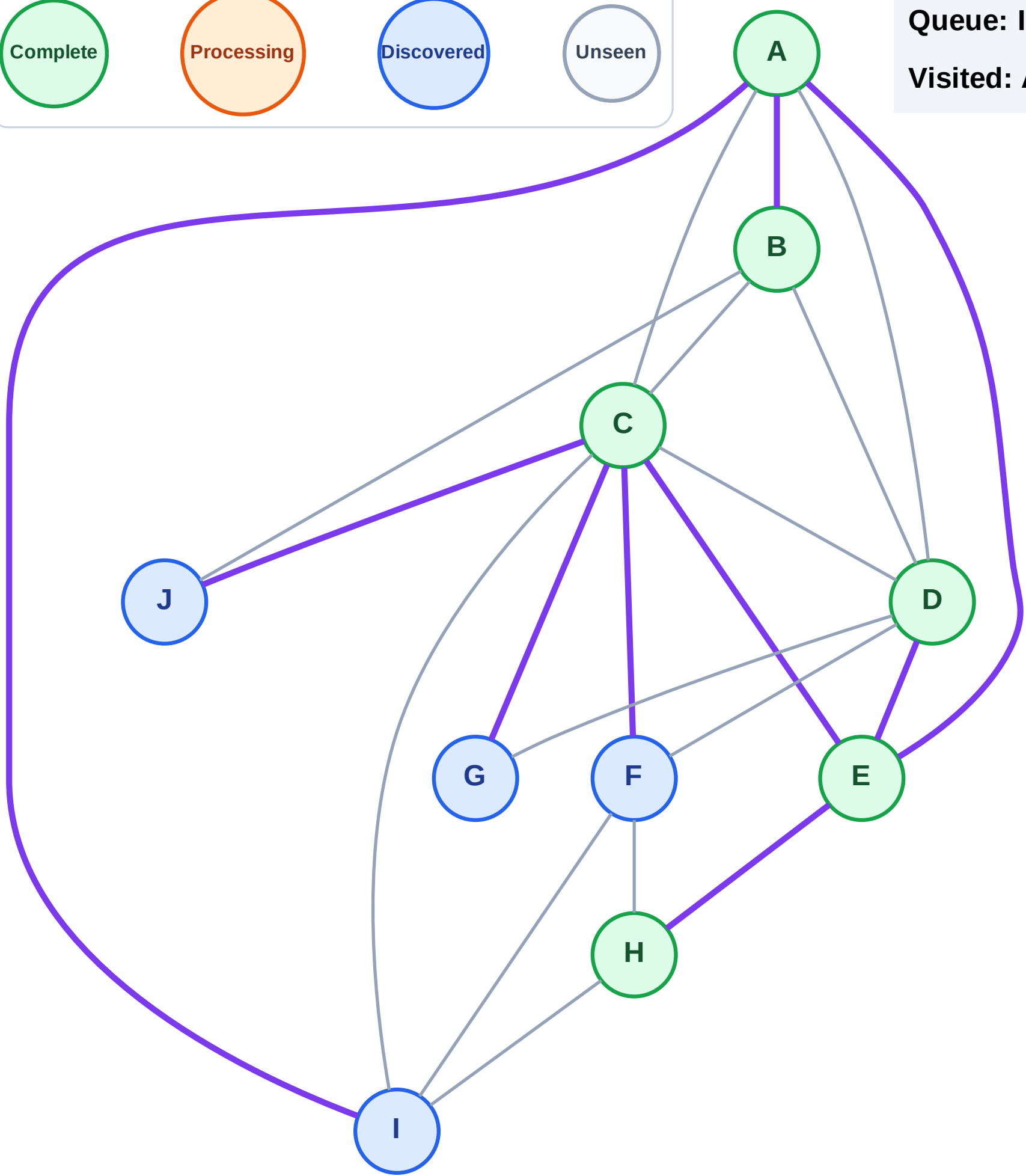
Step 13: No new neighbors from B

Legend



Queue: I → F → G → J

Visited: A, B, C, D, E, H



BFS Traversal

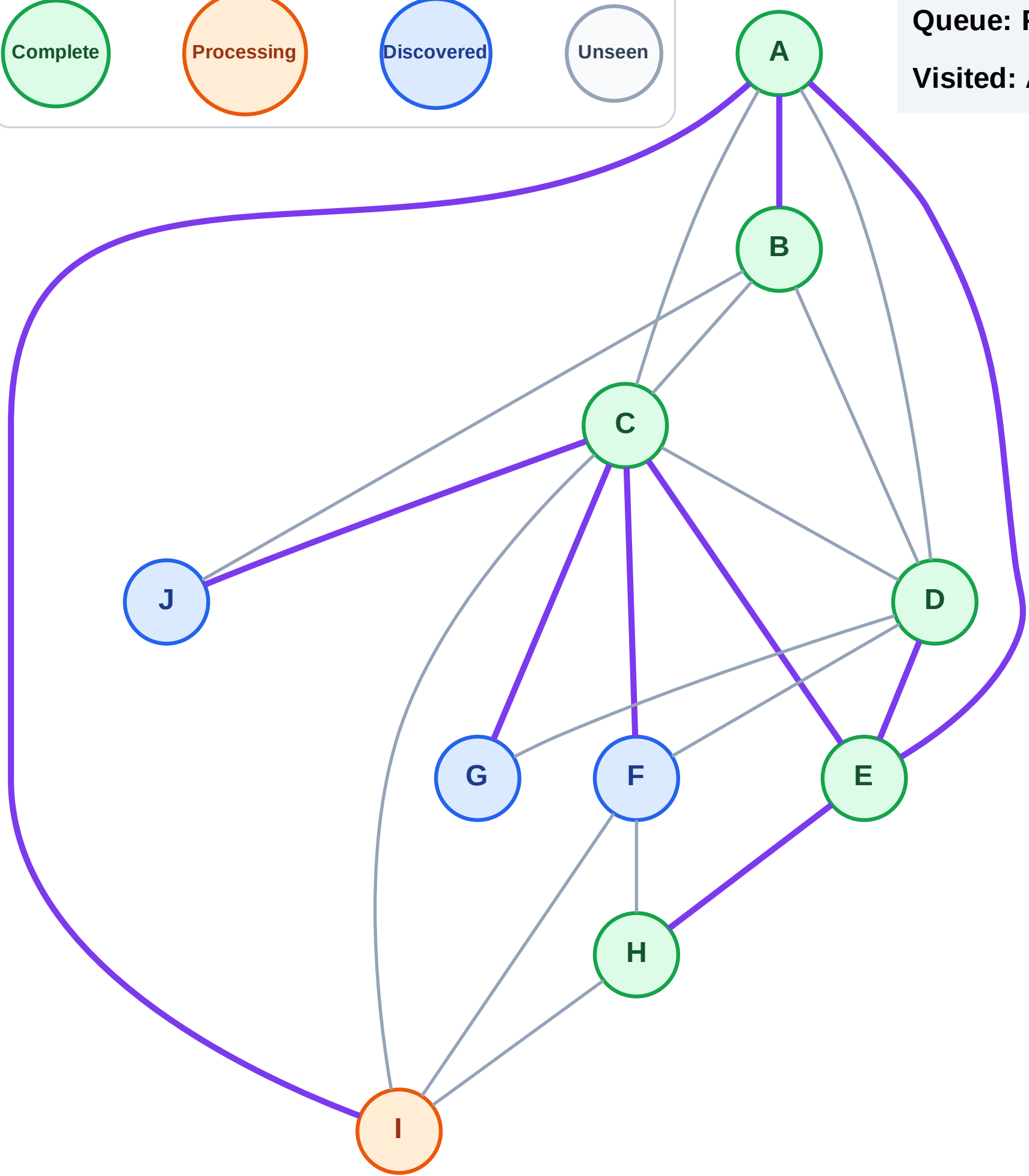
Step 14: Process node I

Legend



Queue: F → G → J

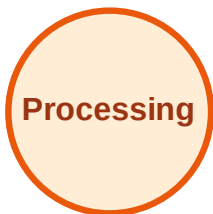
Visited: A, B, C, D, E, H



BFS Traversal

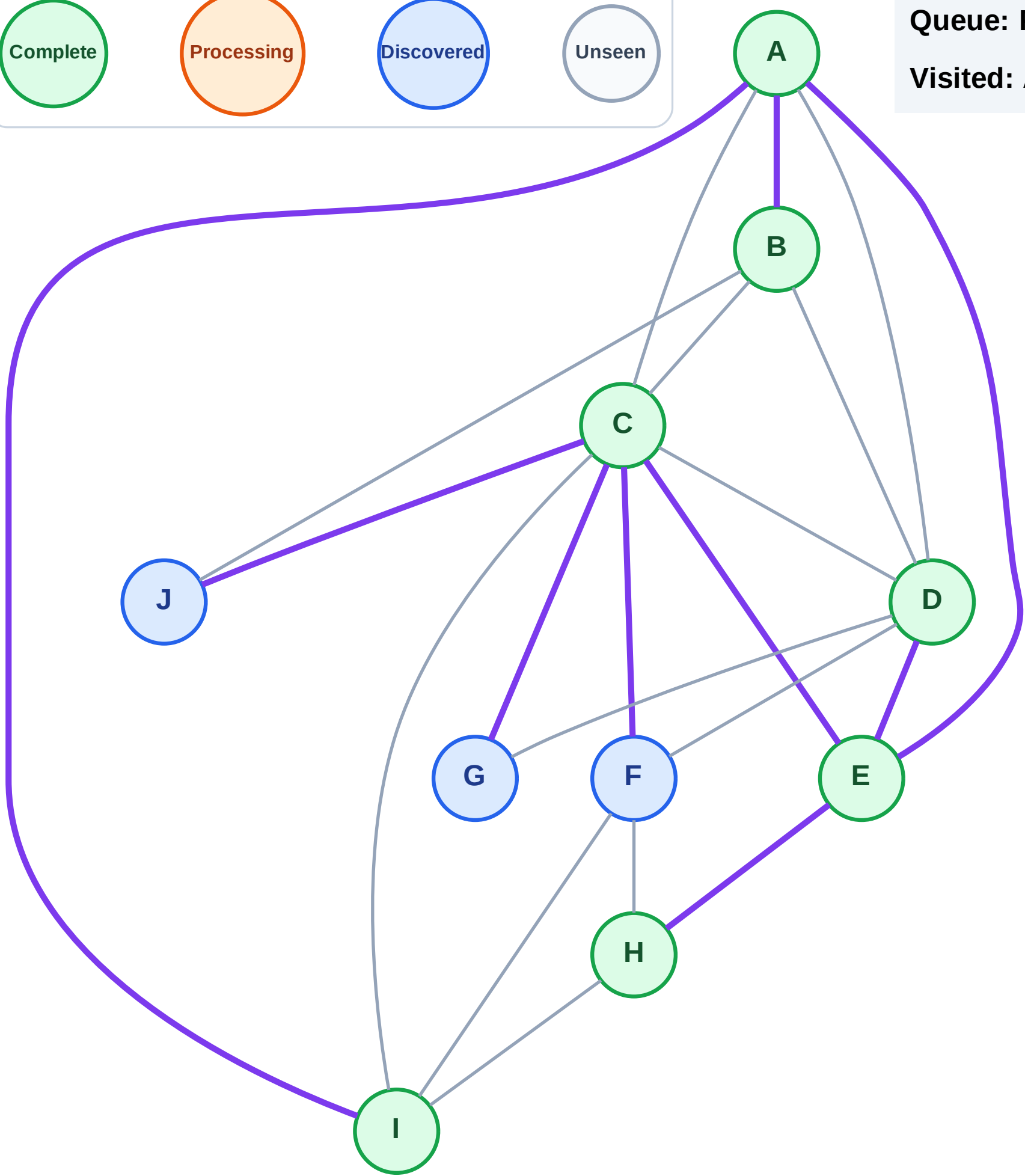
Step 15: No new neighbors from I

Legend



Queue: F → G → J

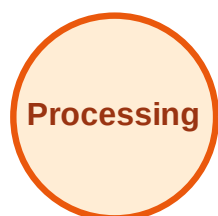
Visited: A, B, C, D, E, H, I



BFS Traversal

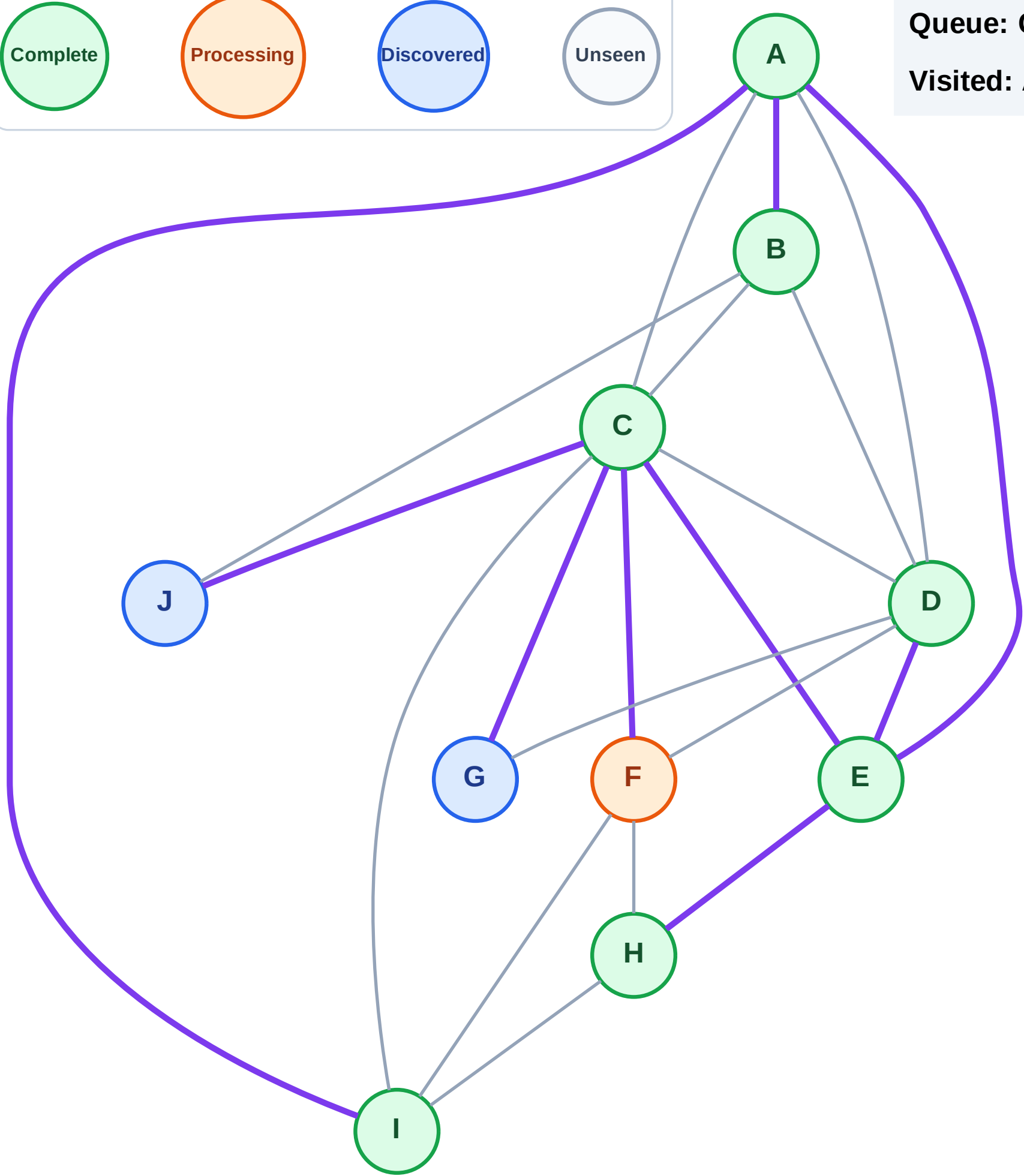
Step 16: Process node F

Legend



Queue: G → J

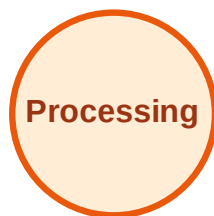
Visited: A, B, C, D, E, H, I



BFS Traversal

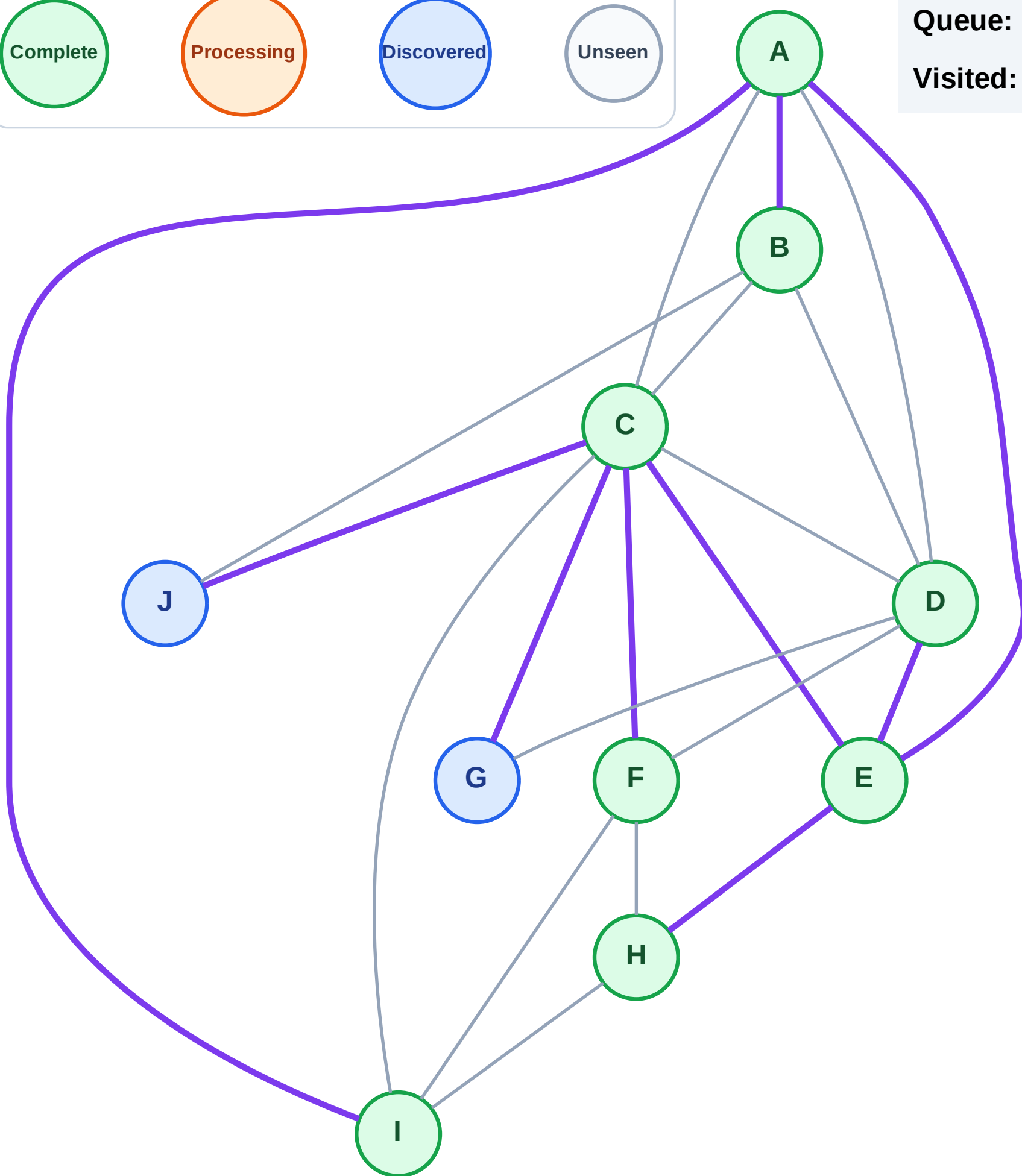
Step 17: No new neighbors from F

Legend



Queue: G → J

Visited: A, B, C, D, E, F, H, I



BFS Traversal

Step 18: Process node G

Legend

Complete

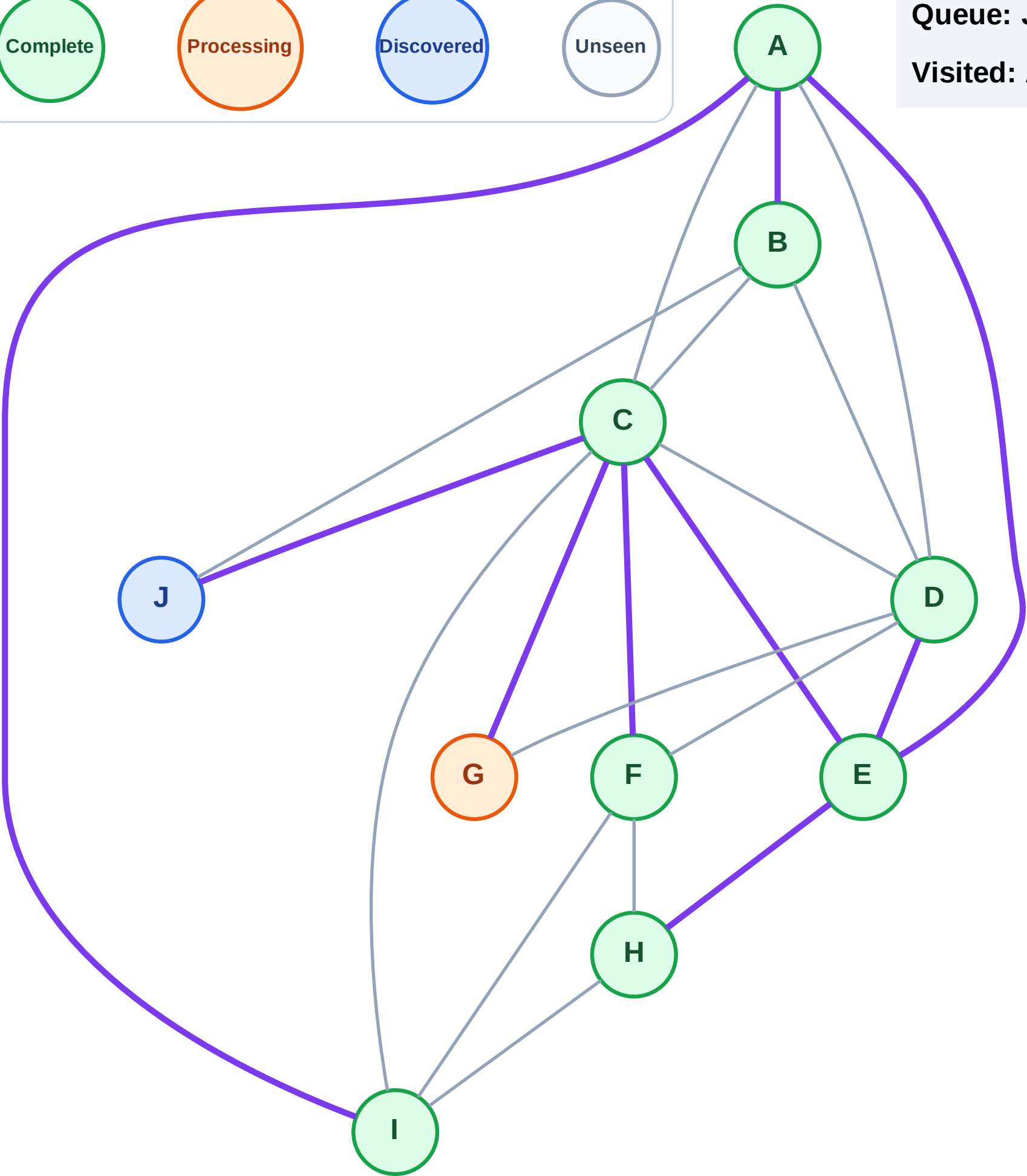
Processing

Discovered

Unseen

Queue: J

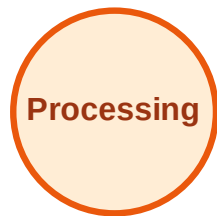
Visited: A, B, C, D, E, F, H, I



BFS Traversal

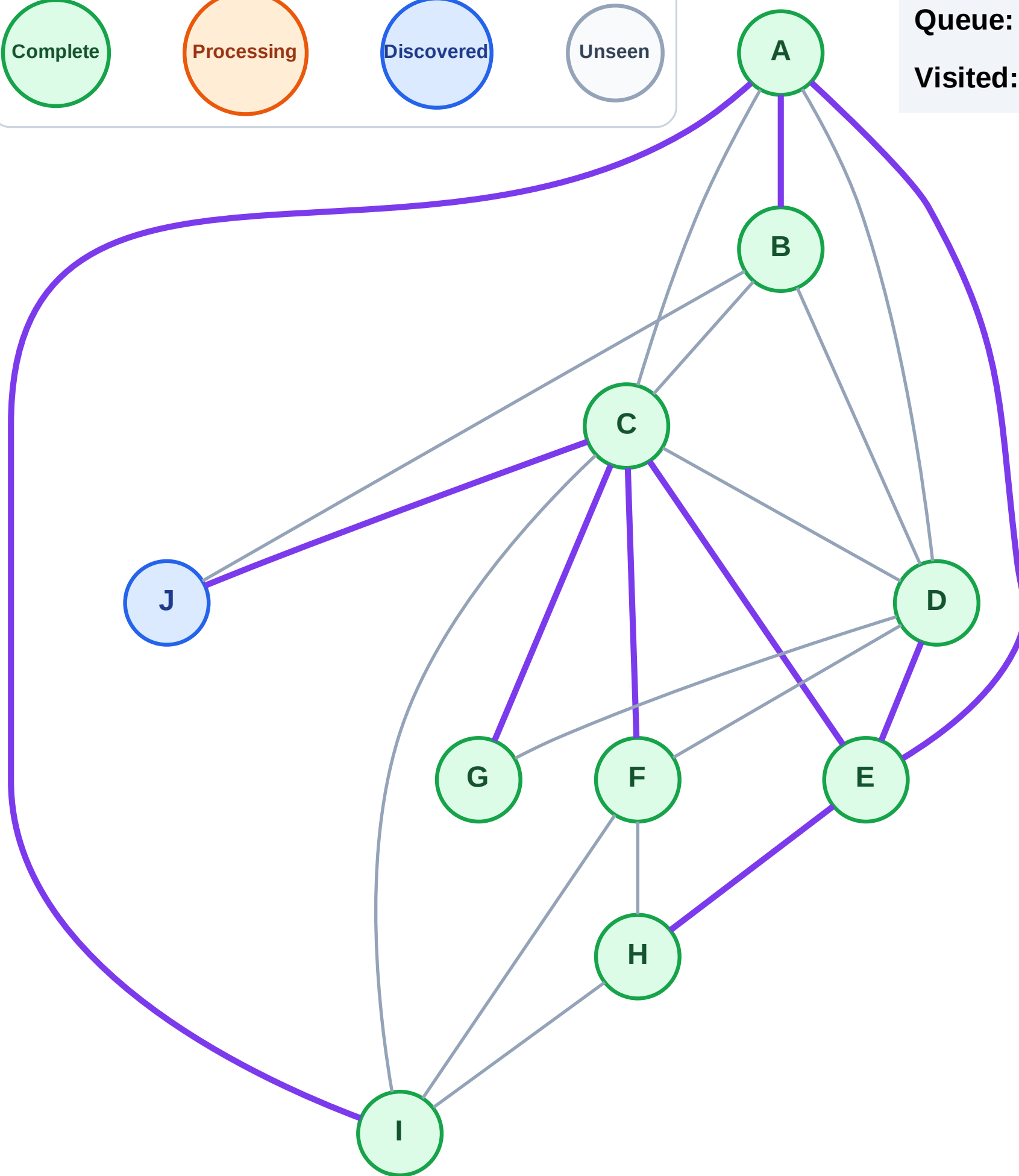
Step 19: No new neighbors from G

Legend



Queue: J

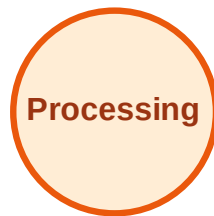
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

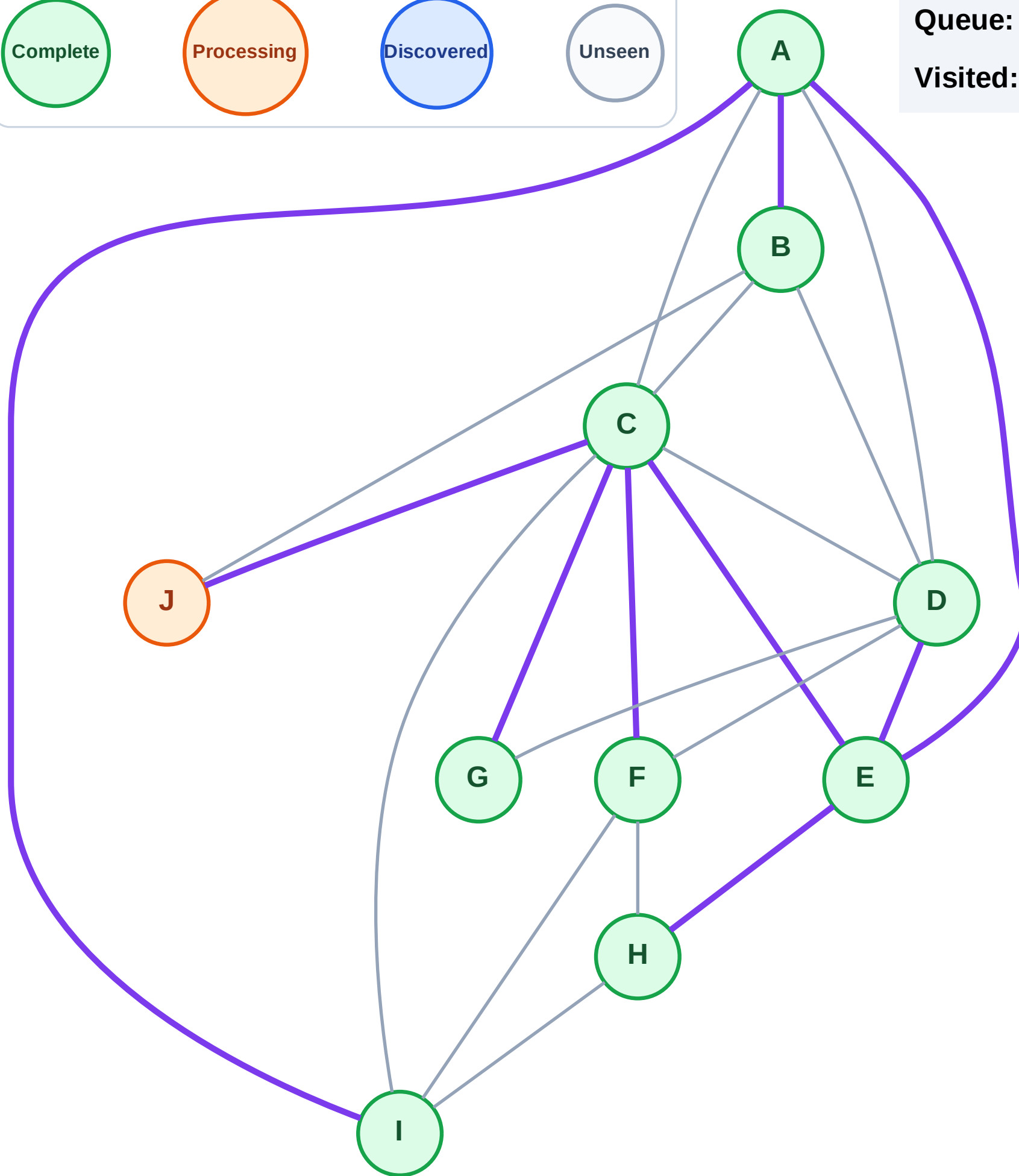
Step 20: Process node J

Legend



Queue: (empty)

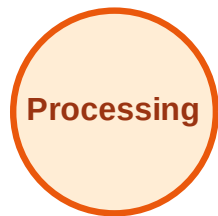
Visited: A, B, C, D, E, F, G, H, I



BFS Traversal

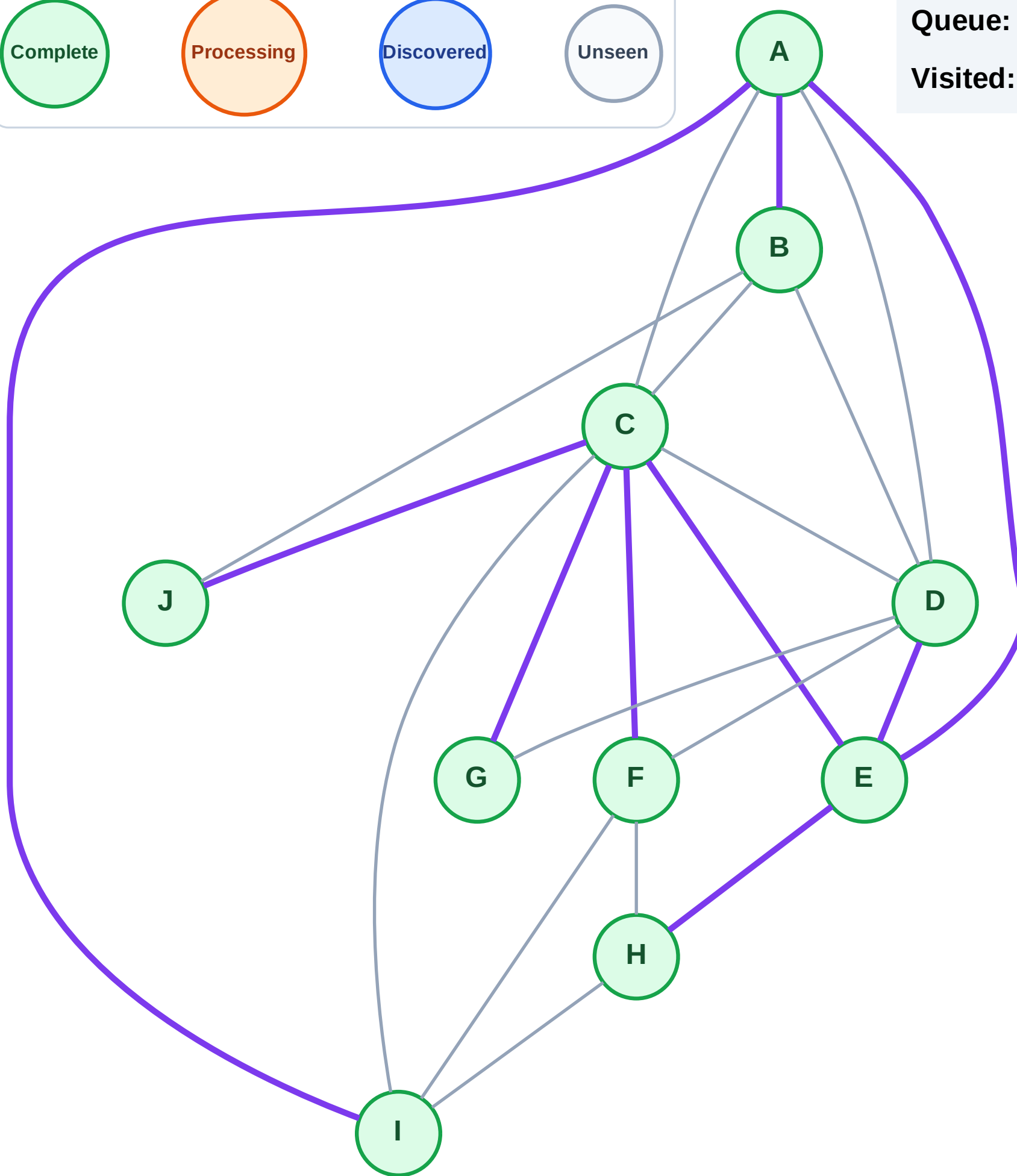
Step 21: No new neighbors from J

Legend



Queue: (empty)

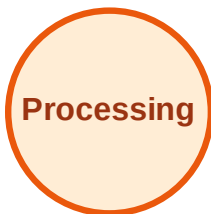
Visited: A, B, C, D, E, F, G, H, I, J



DFS Traversal

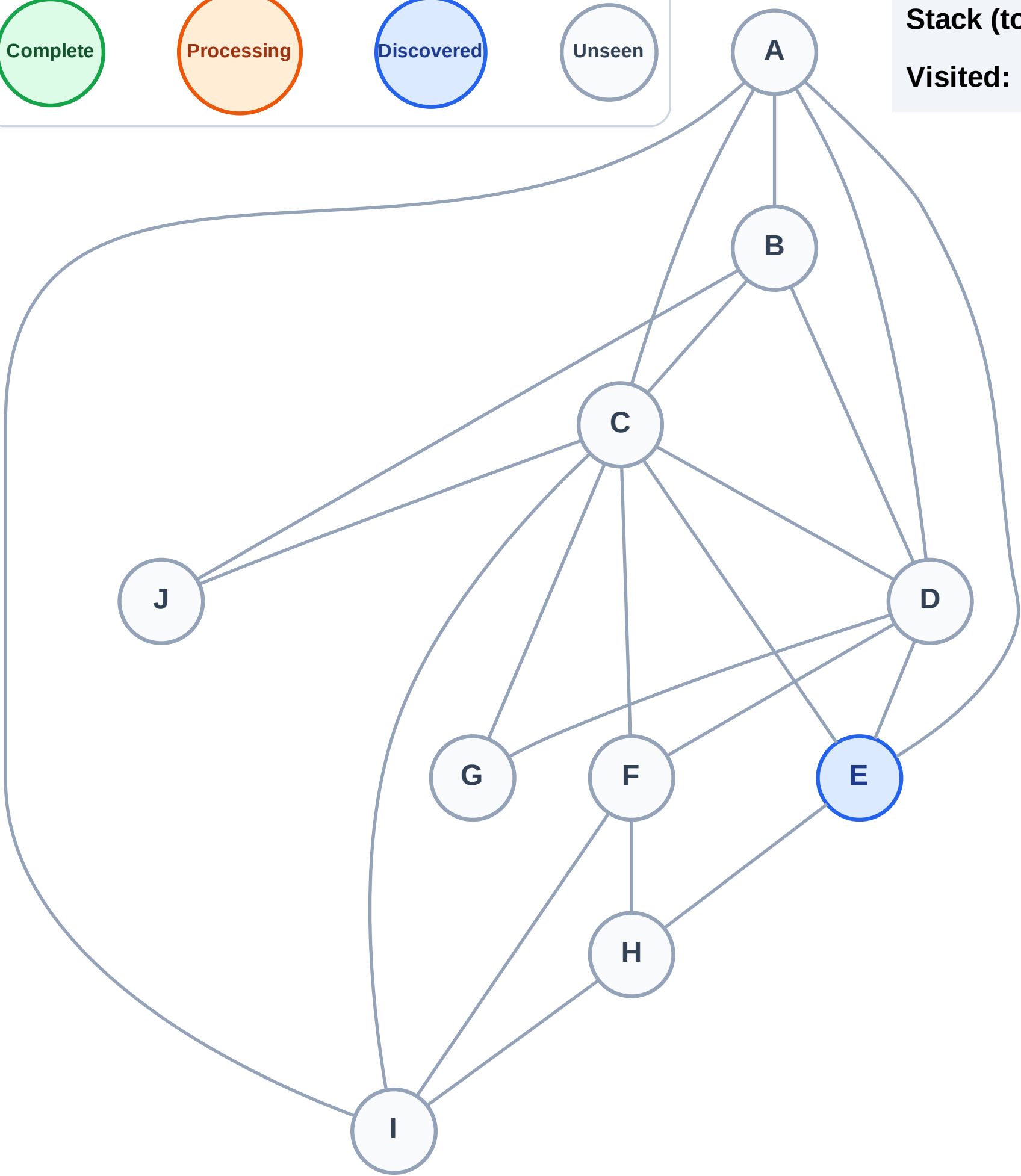
Step 1: Start at E

Legend



Stack (top at right): E

Visited: (none)



DFS Traversal

Step 2: Process node E

Legend

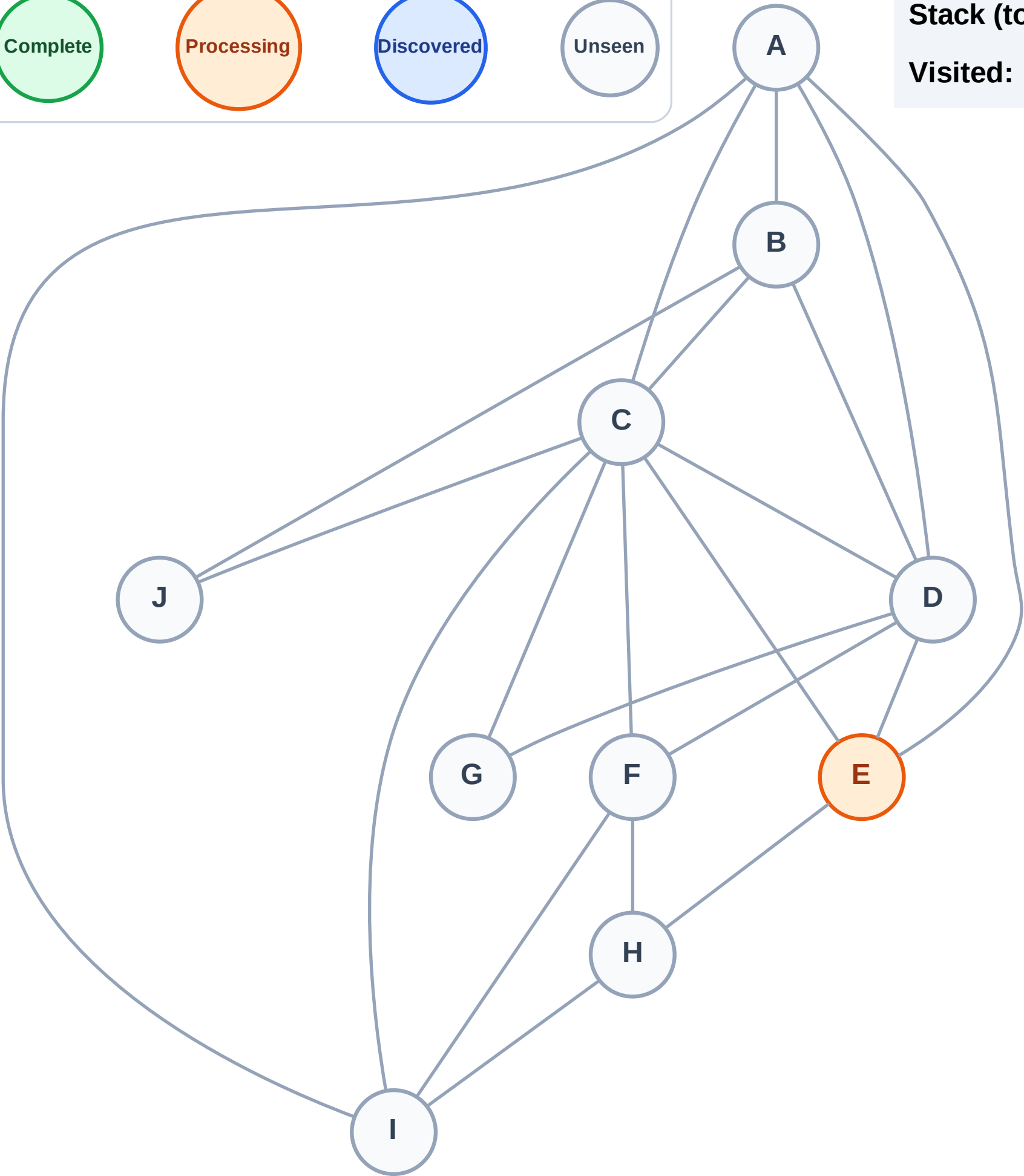
Complete

Processing

Discovered

Unseen

Stack (top at right): (empty)
Visited: (none)



DFS Traversal

Step 3: Discover H, D, C, A from E

Legend

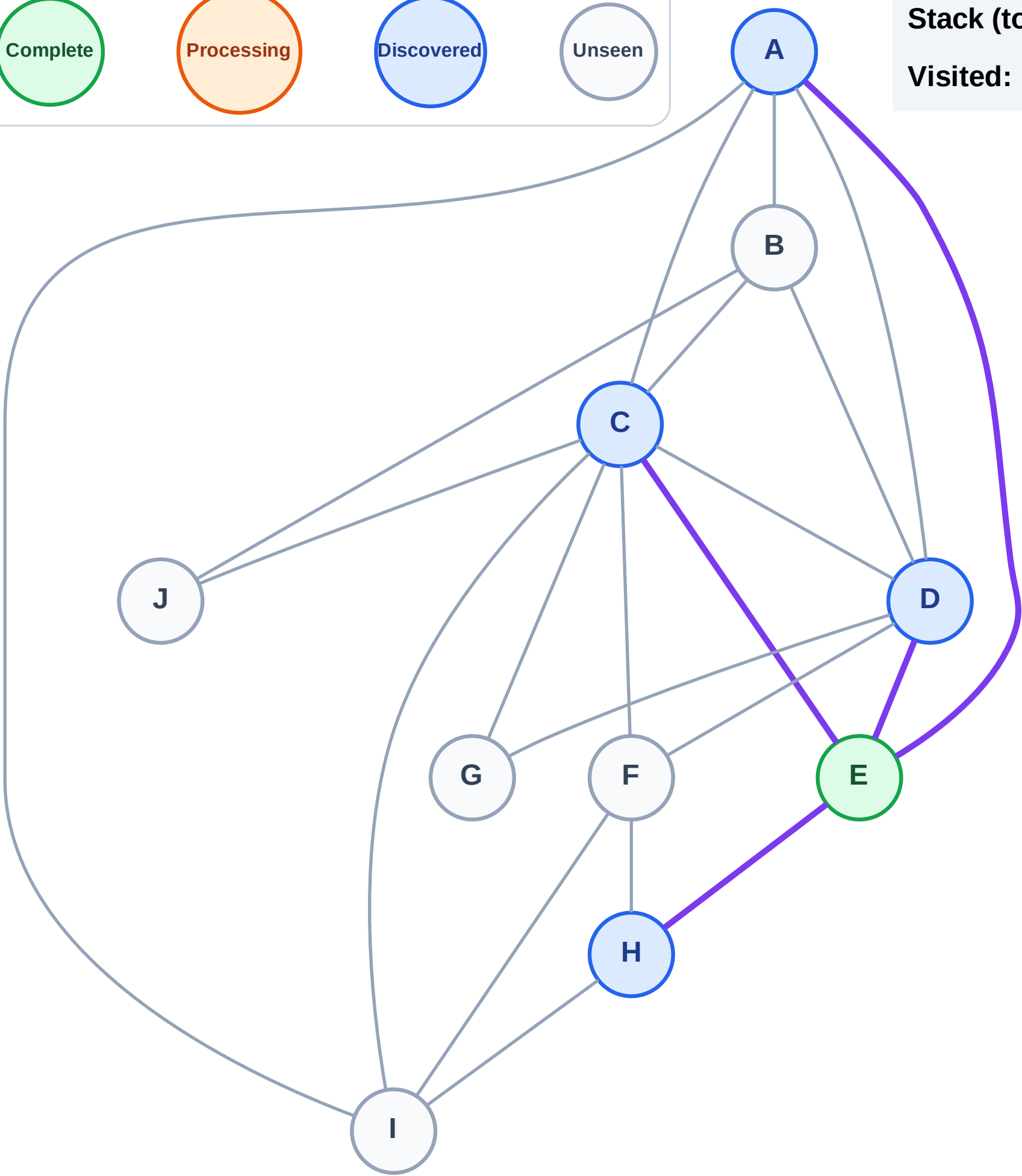
Complete

Processing

Discovered

Unseen

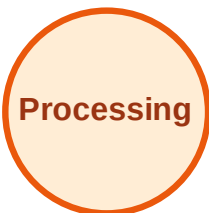
Stack (top at right): H | D | C | A
Visited: E



DFS Traversal

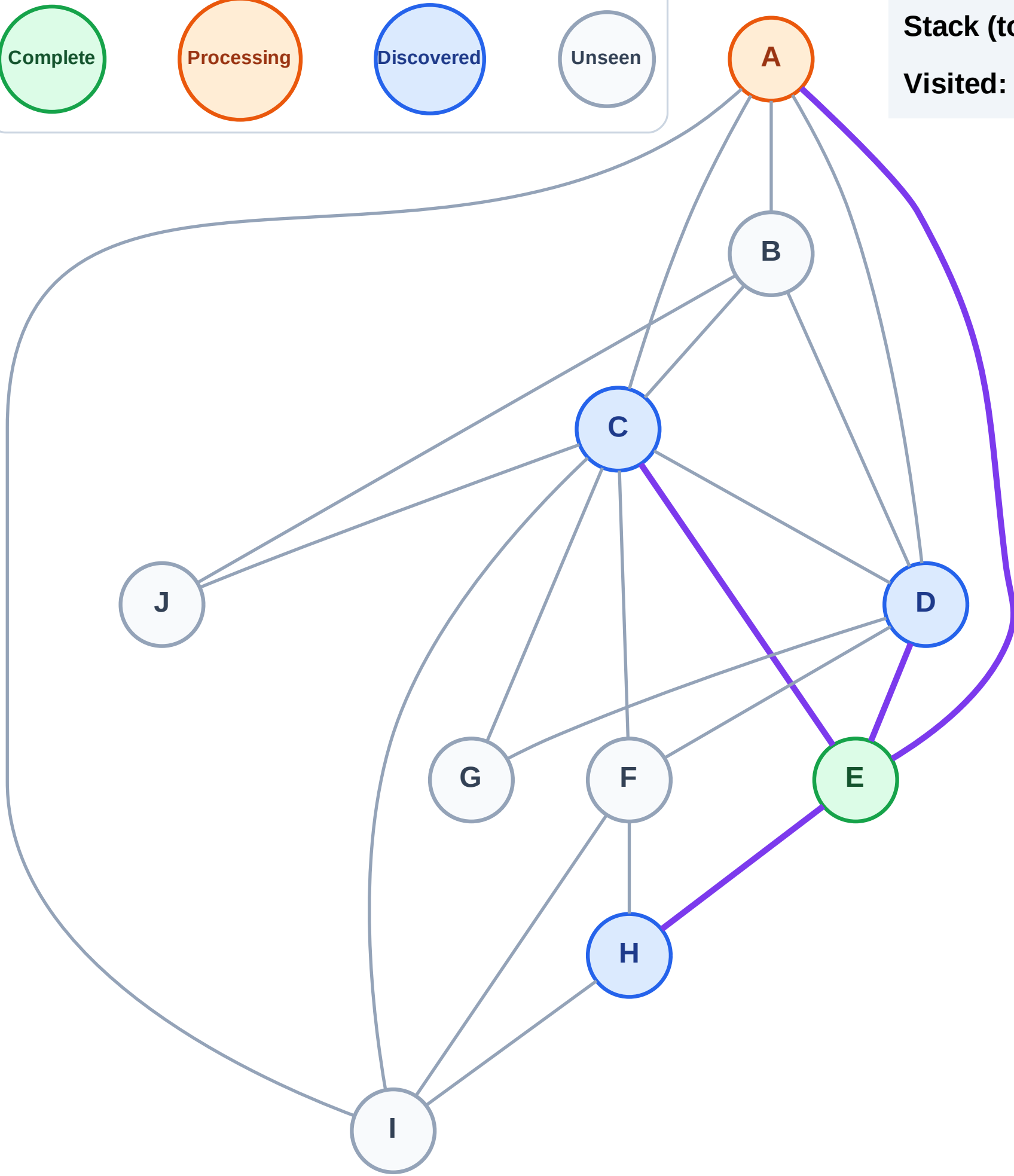
Step 4: Process node A

Legend



Stack (top at right): H | D | C

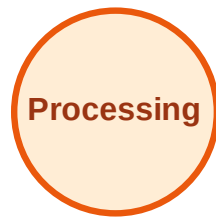
Visited: E



DFS Traversal

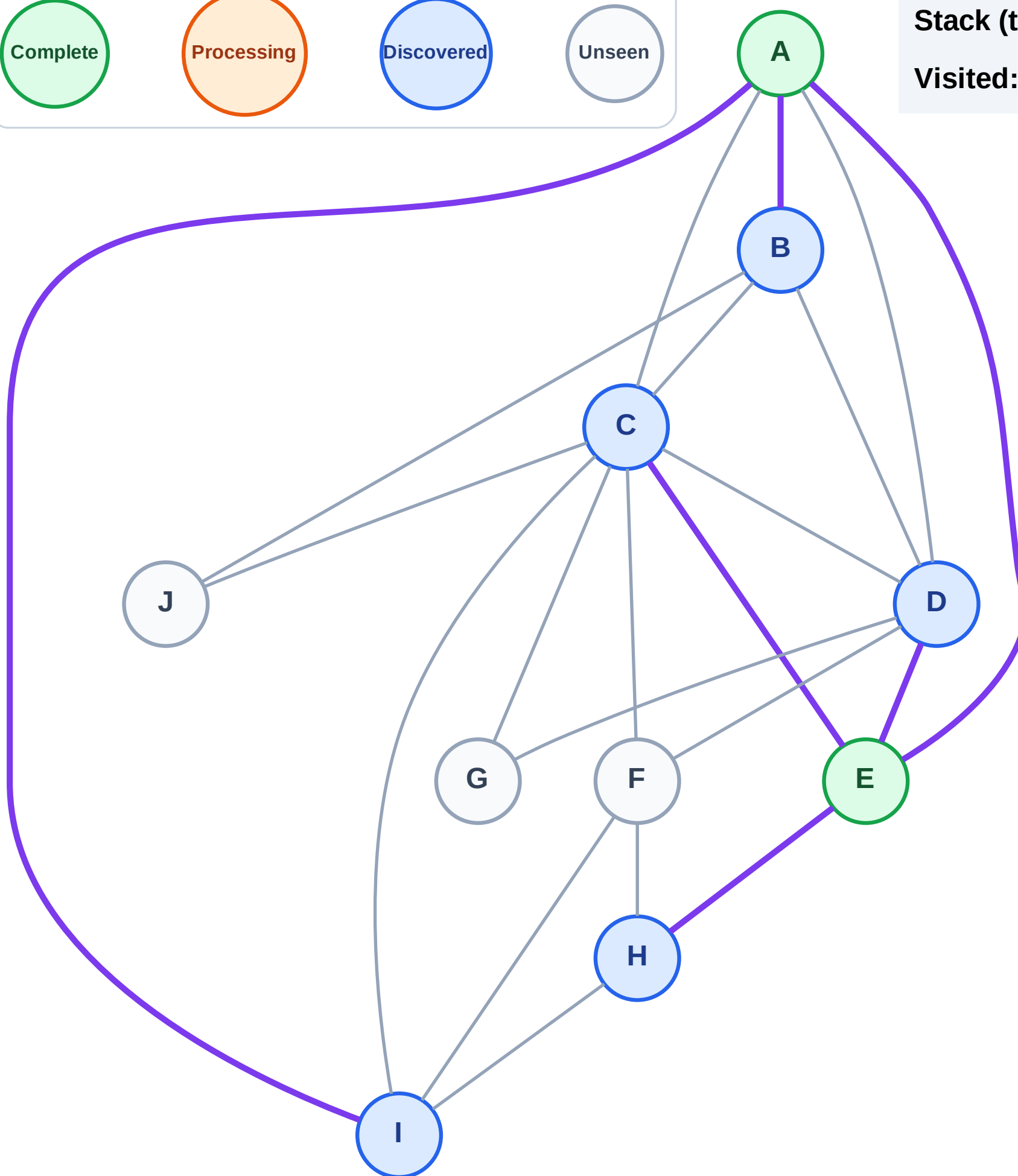
Step 5: Discover I, B from A

Legend



Stack (top at right): H | D | C | I | B

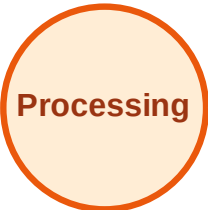
Visited: A, E



DFS Traversal

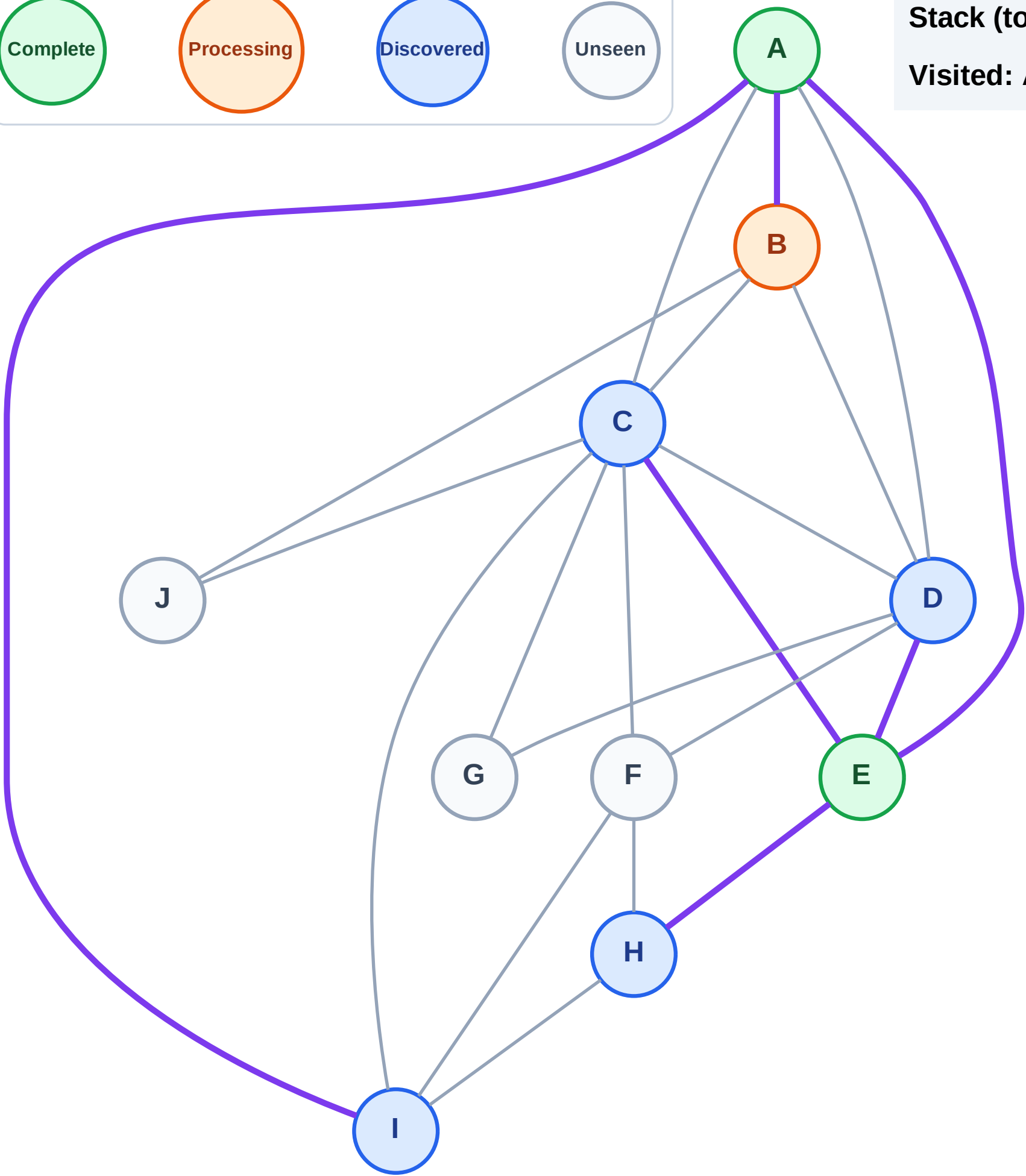
Step 6: Process node B

Legend



Stack (top at right): H | D | C | I

Visited: A, E



DFS Traversal

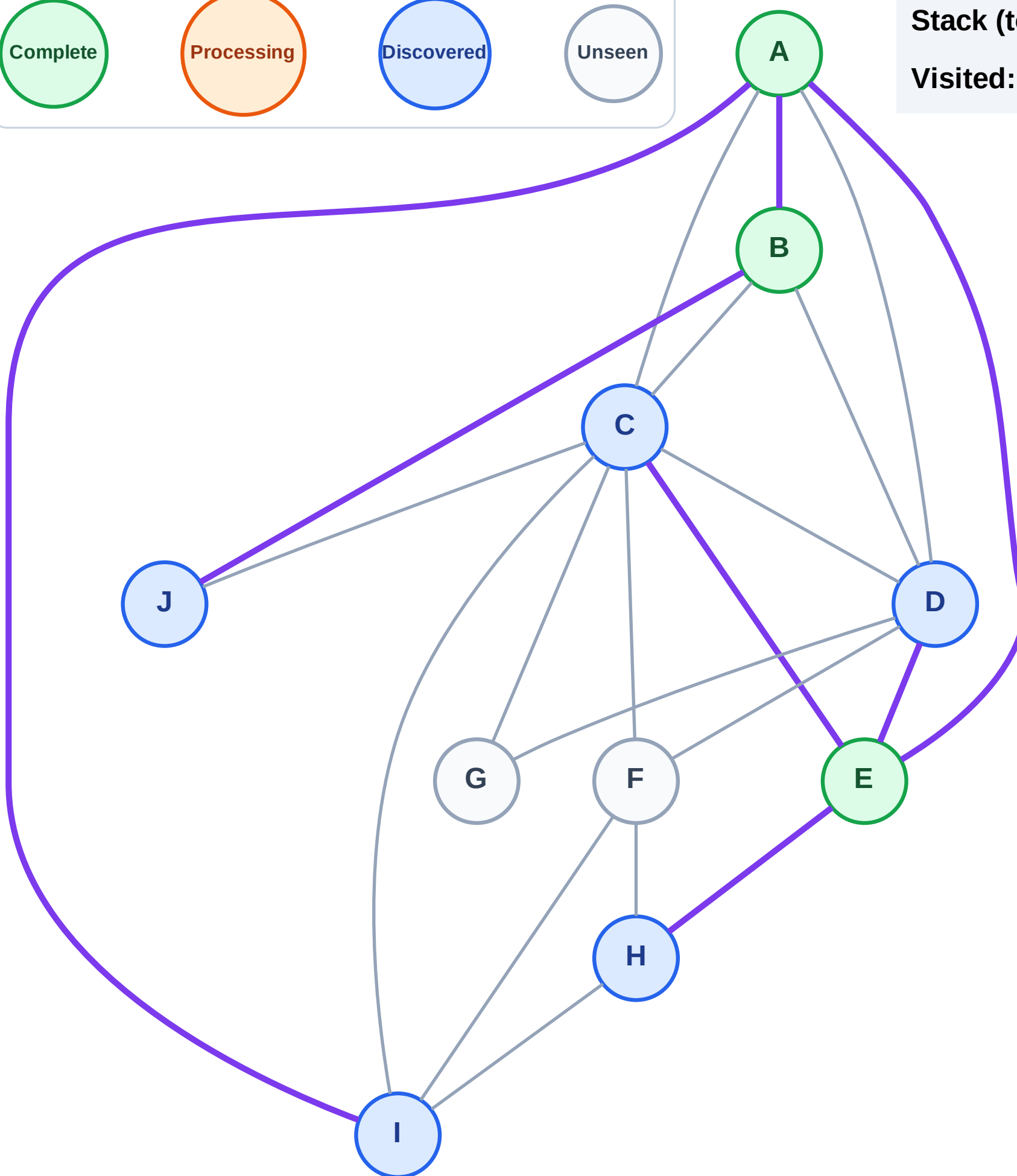
Step 7: Discover J from B

Legend



Stack (top at right): H | D | C | I | J

Visited: A, B, E



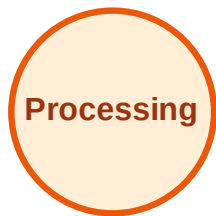
DFS Traversal

Step 8: Process node J

Legend



Complete



Processing



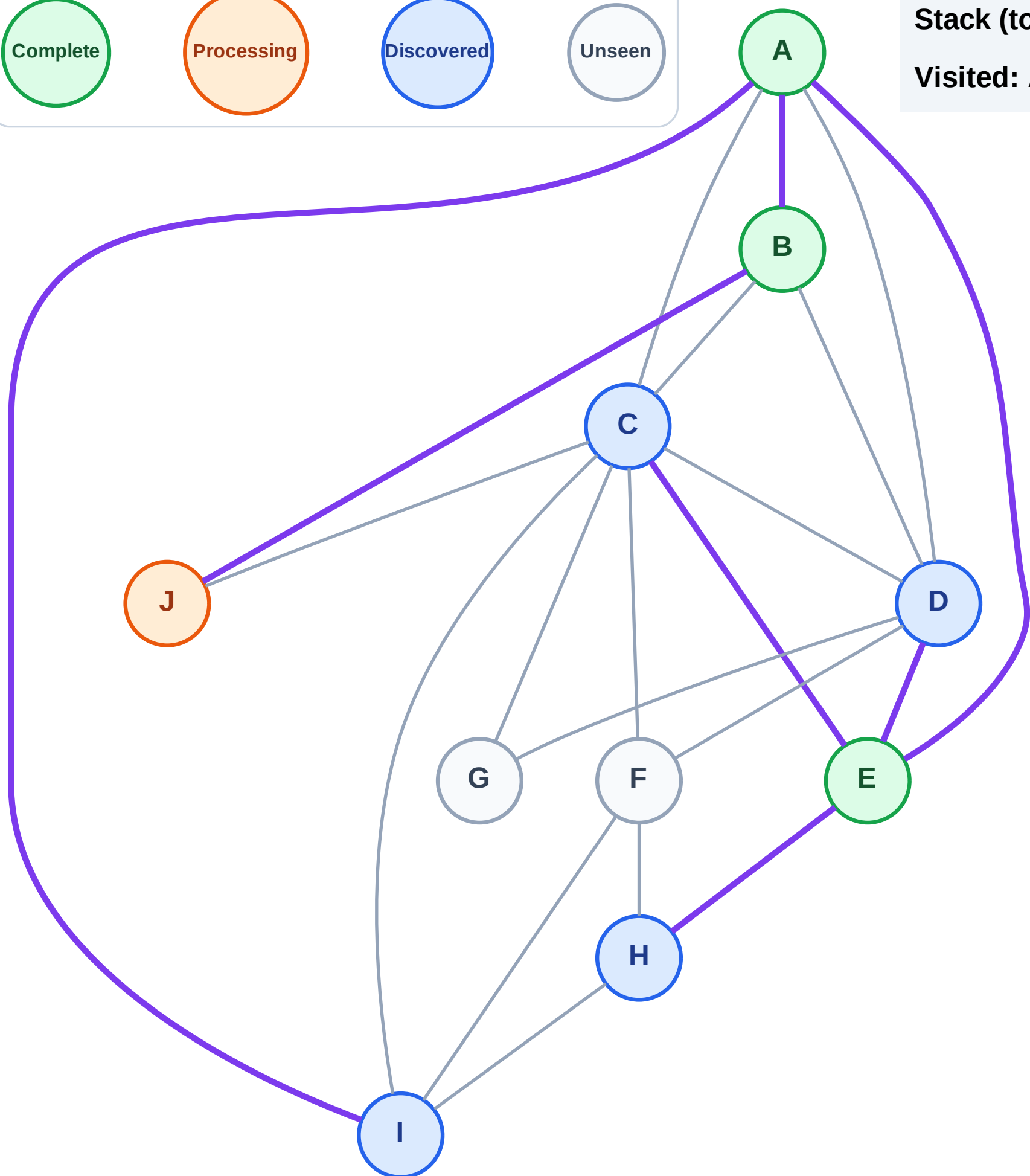
Discovered



Unseen

Stack (top at right): H | D | C | I

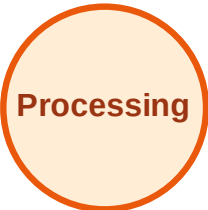
Visited: A, B, E



DFS Traversal

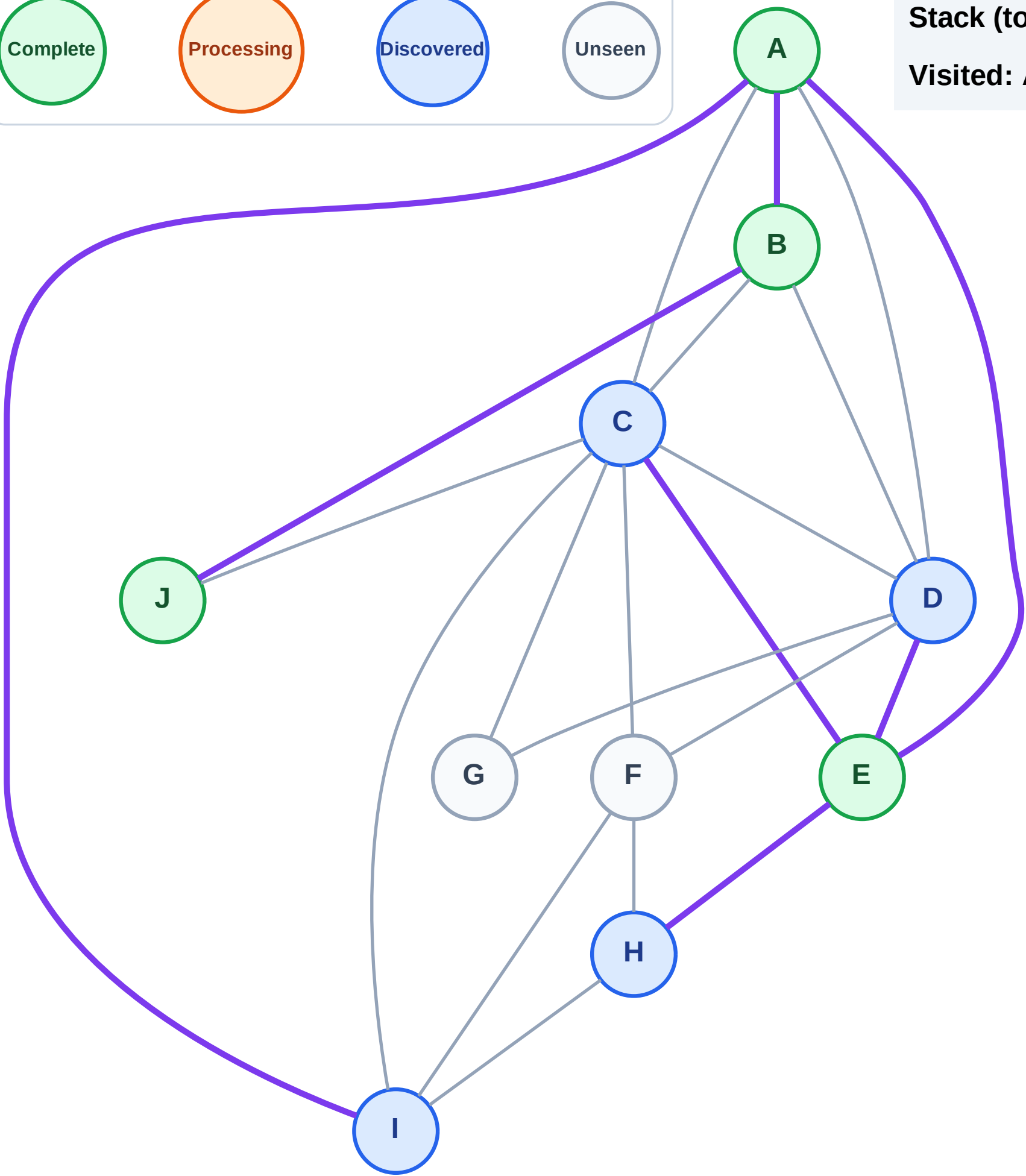
Step 9: No new neighbors from J

Legend



Stack (top at right): H | D | C | I

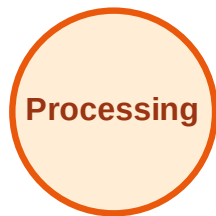
Visited: A, B, E, J



DFS Traversal

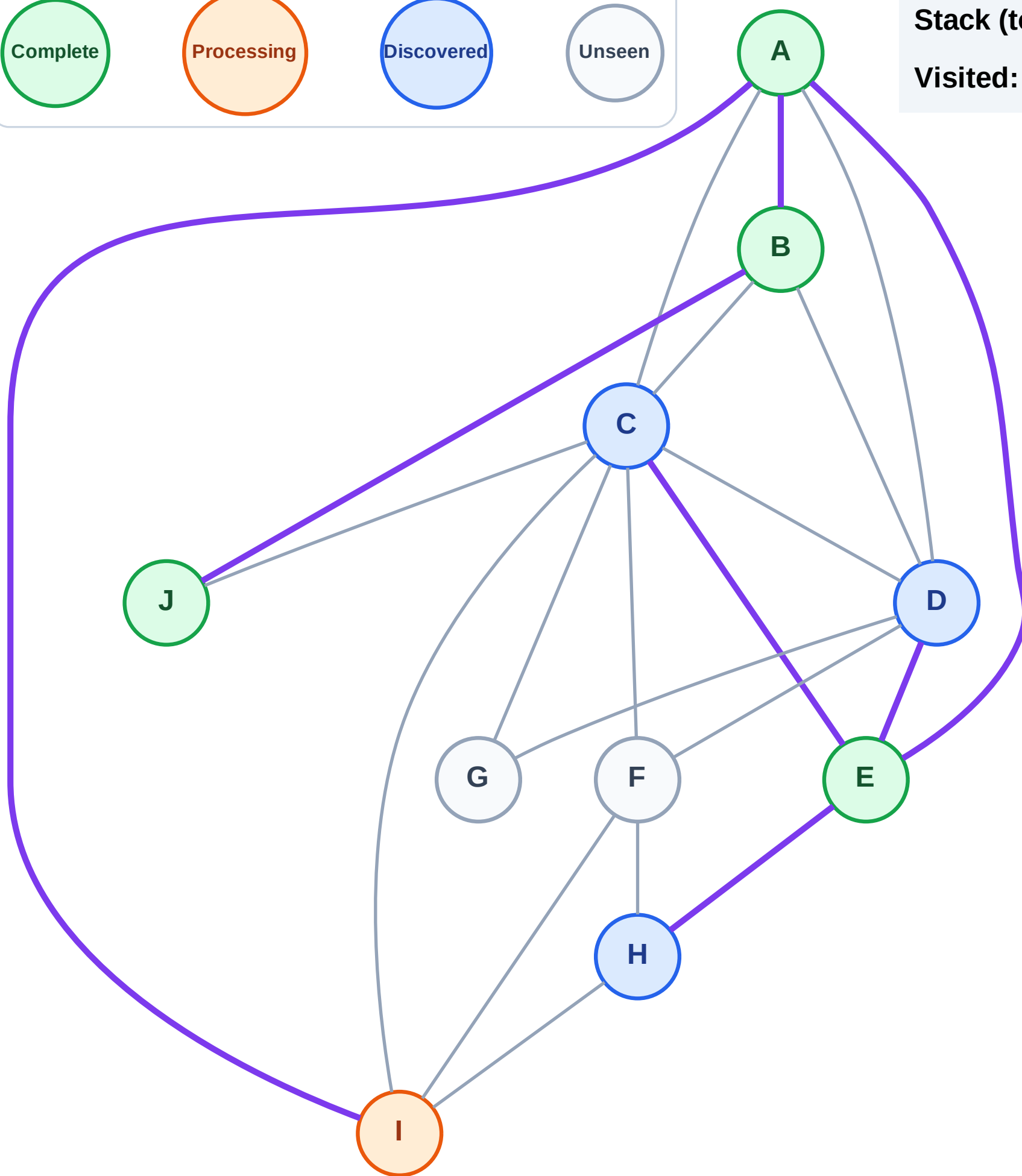
Step 10: Process node I

Legend



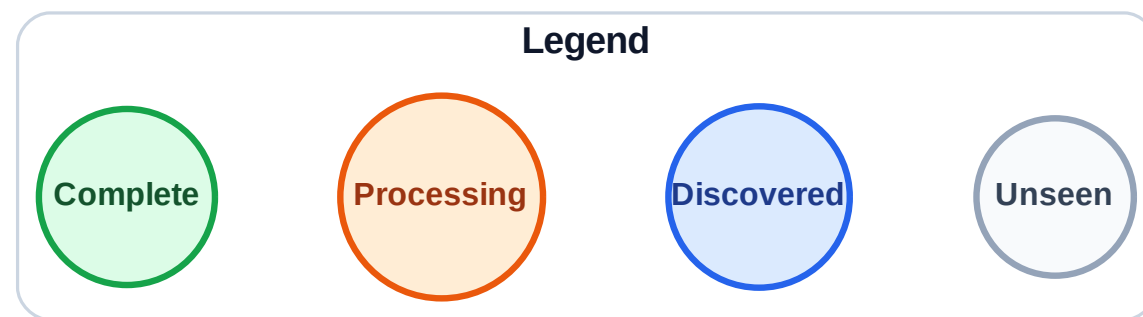
Stack (top at right): H | D | C

Visited: A, B, E, J

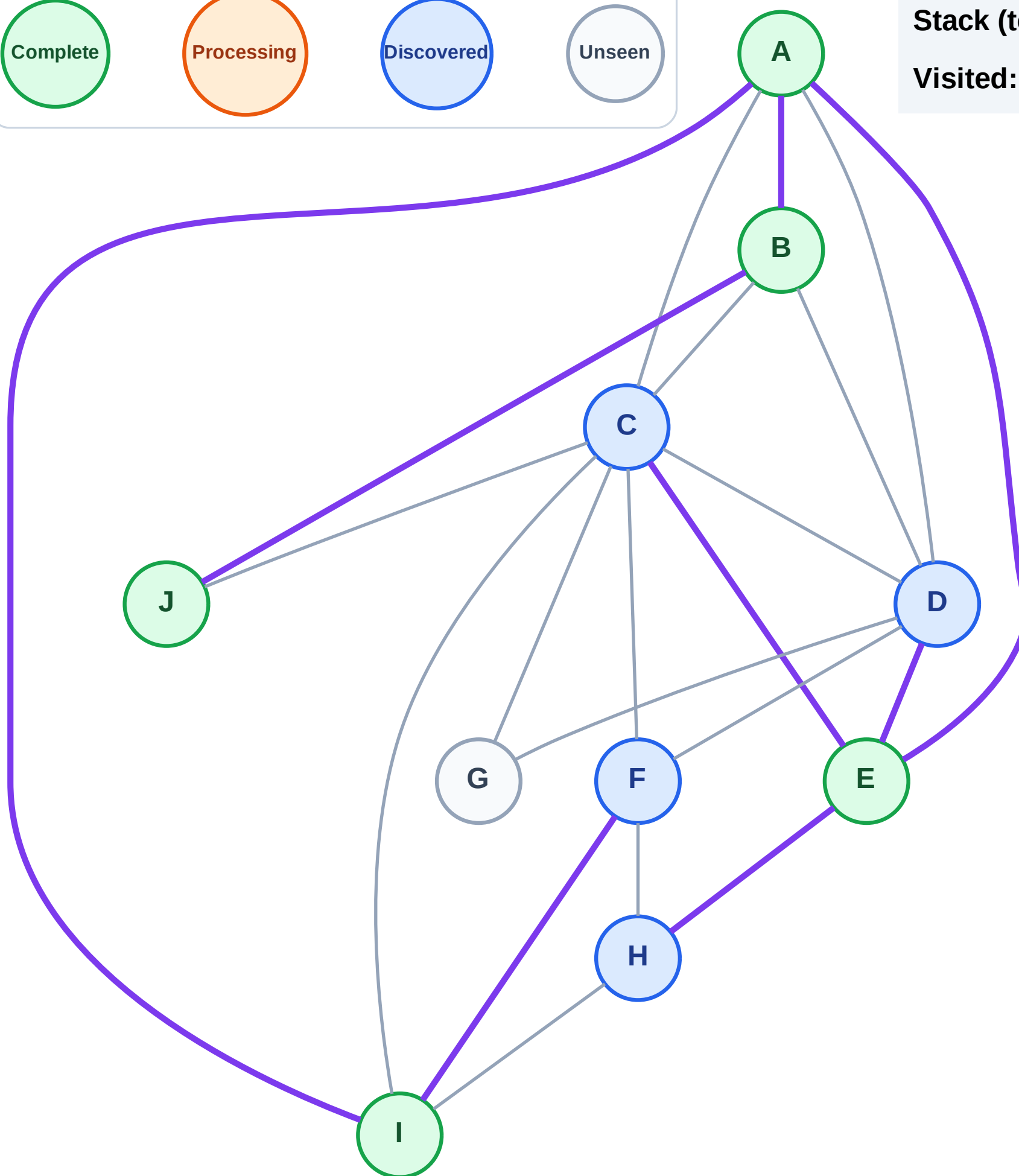


DFS Traversal

Step 11: Discover F from I



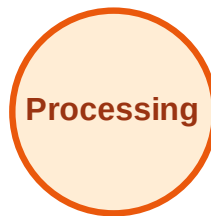
Stack (top at right): H | D | C | F
Visited: A, B, E, I, J



DFS Traversal

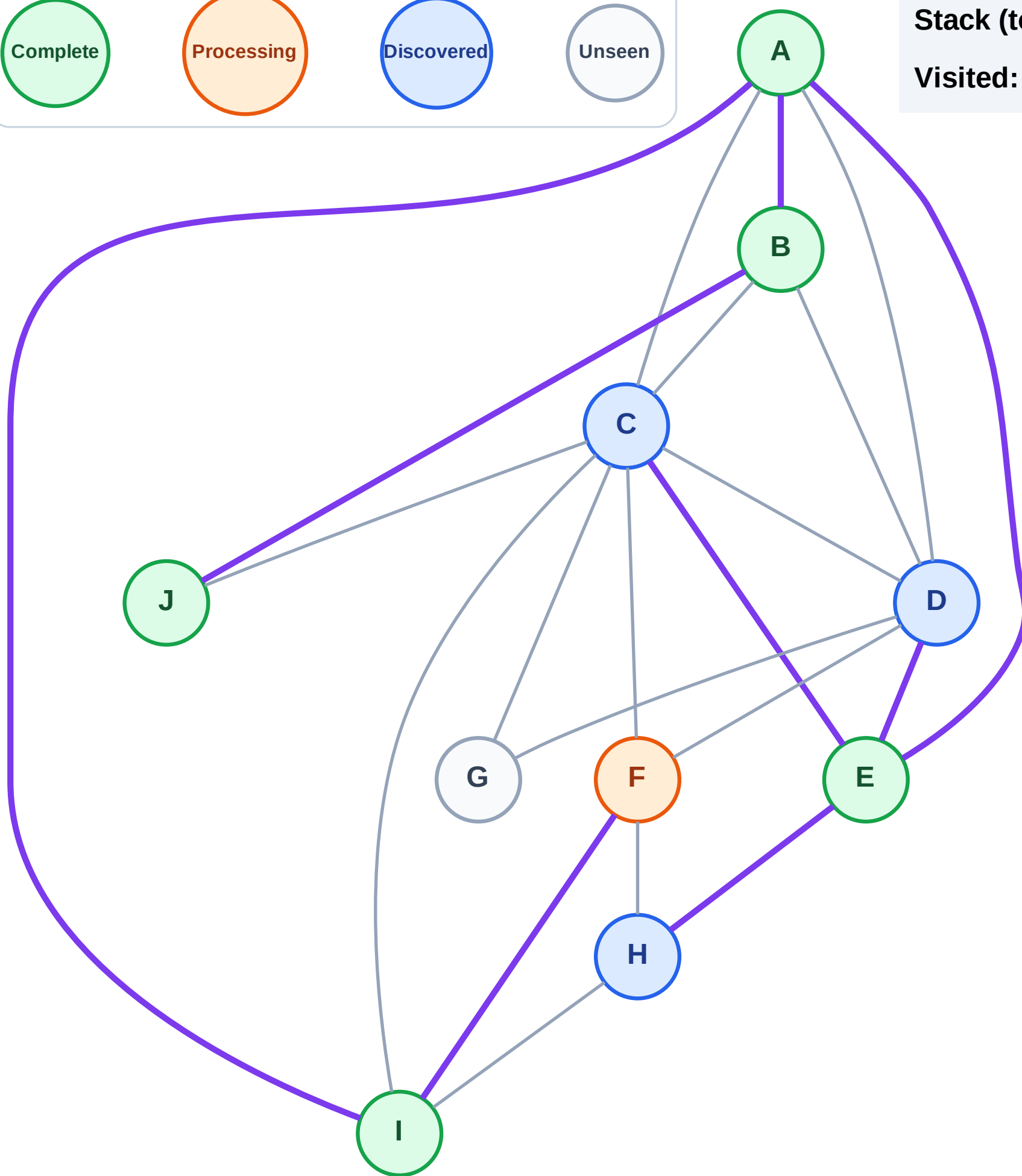
Step 12: Process node F

Legend



Stack (top at right): H | D | C

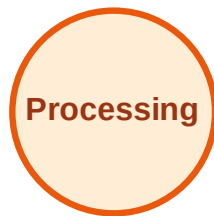
Visited: A, B, E, I, J



DFS Traversal

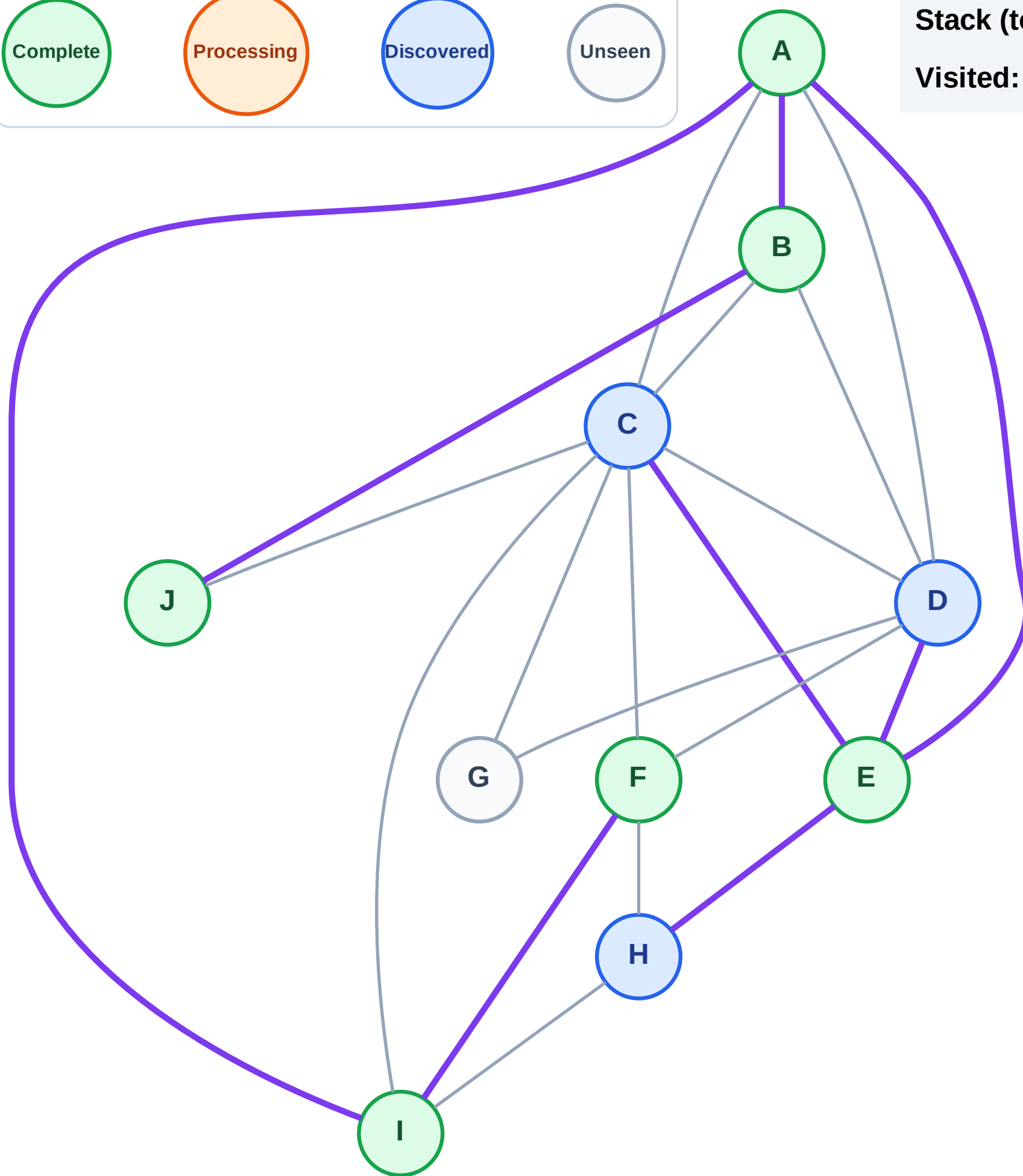
Step 13: No new neighbors from F

Legend



Stack (top at right): H | D | C

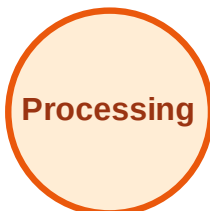
Visited: A, B, E, F, I, J



DFS Traversal

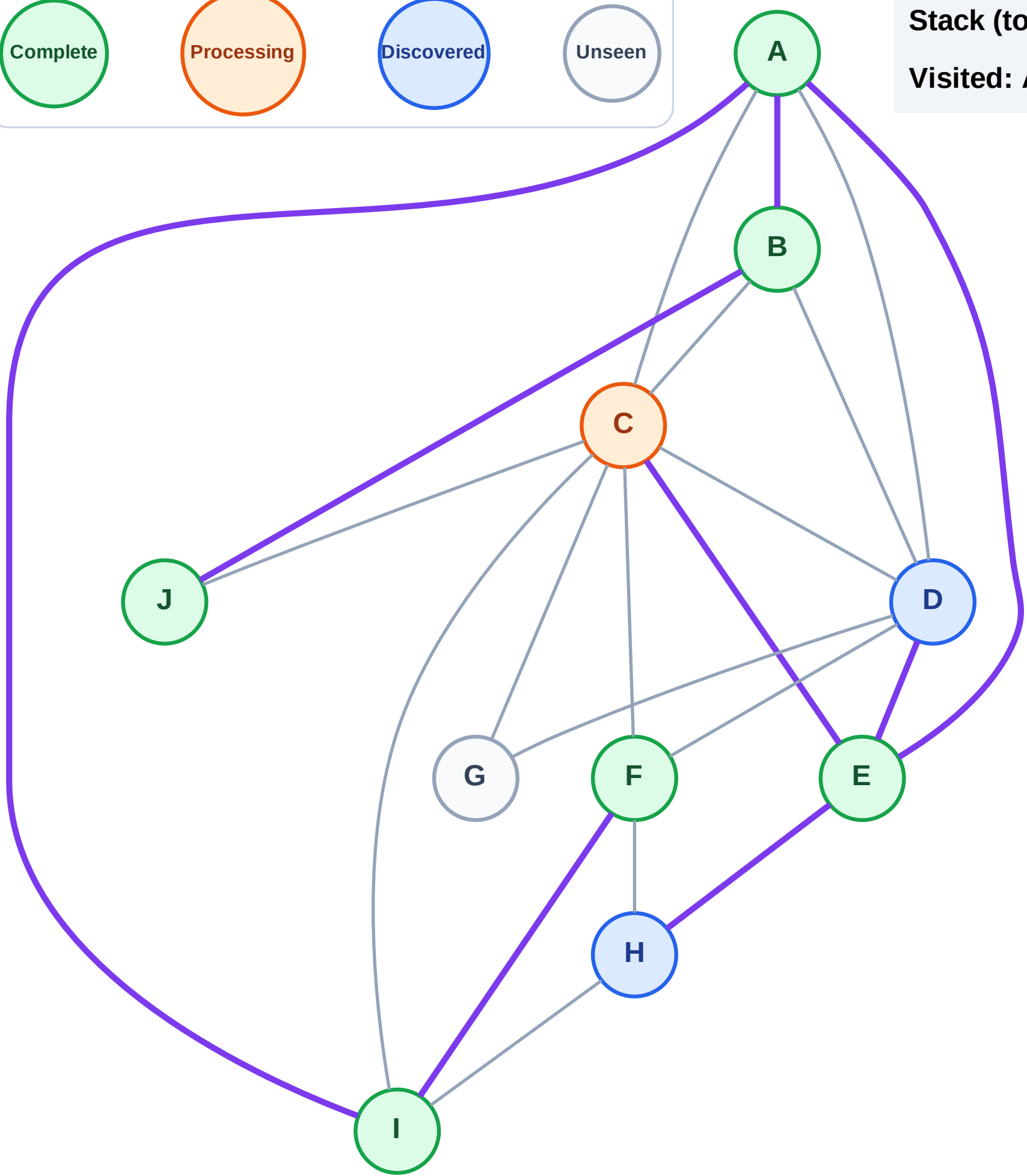
Step 14: Process node C

Legend



Stack (top at right): H | D

Visited: A, B, E, F, I, J



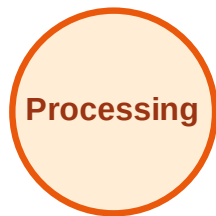
DFS Traversal

Step 15: Discover G from C

Legend



Complete



Processing



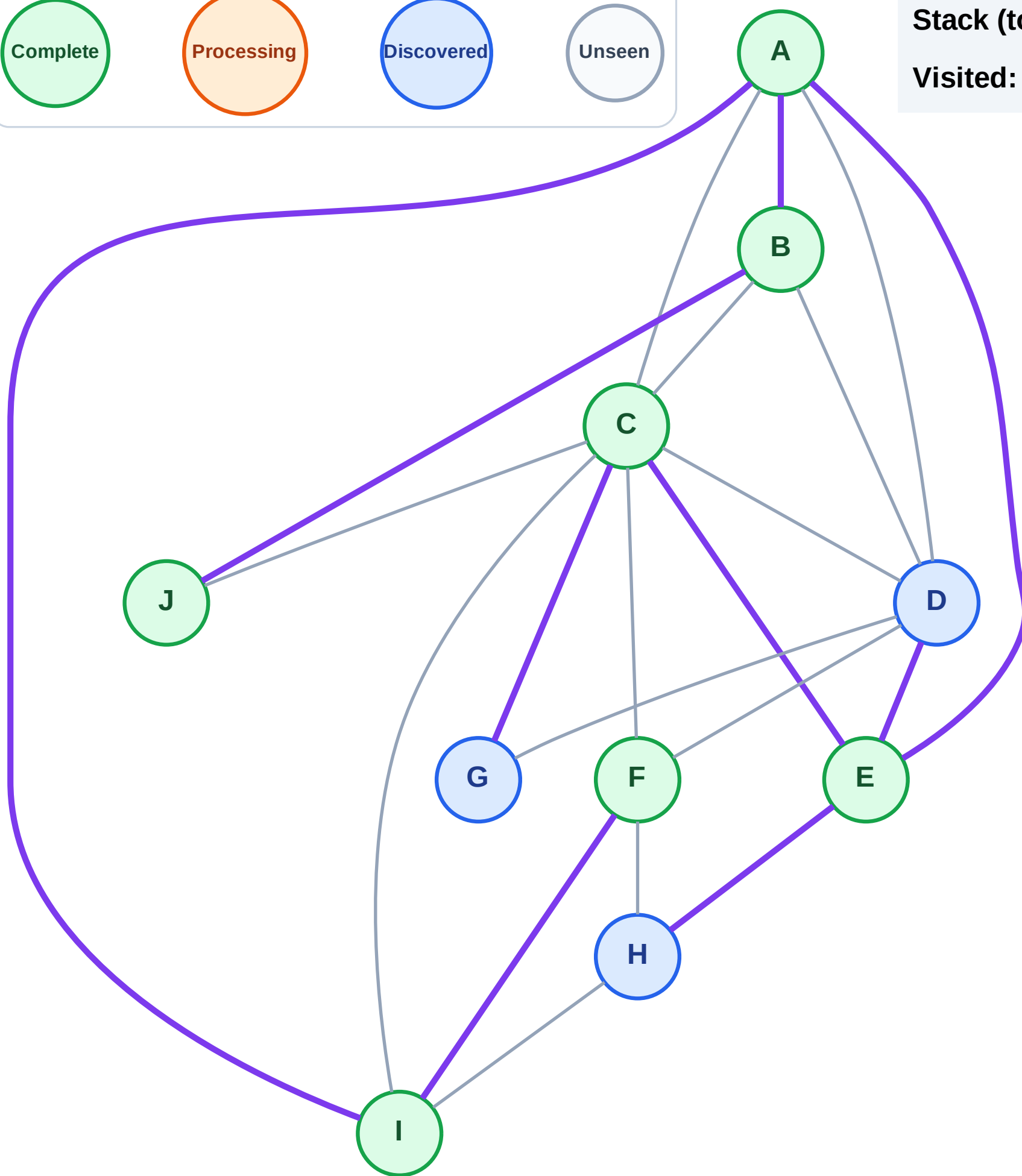
Discovered



Unseen

Stack (top at right): H | D | G

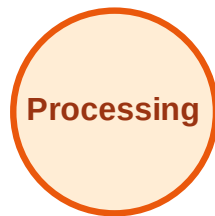
Visited: A, B, C, E, F, I, J



DFS Traversal

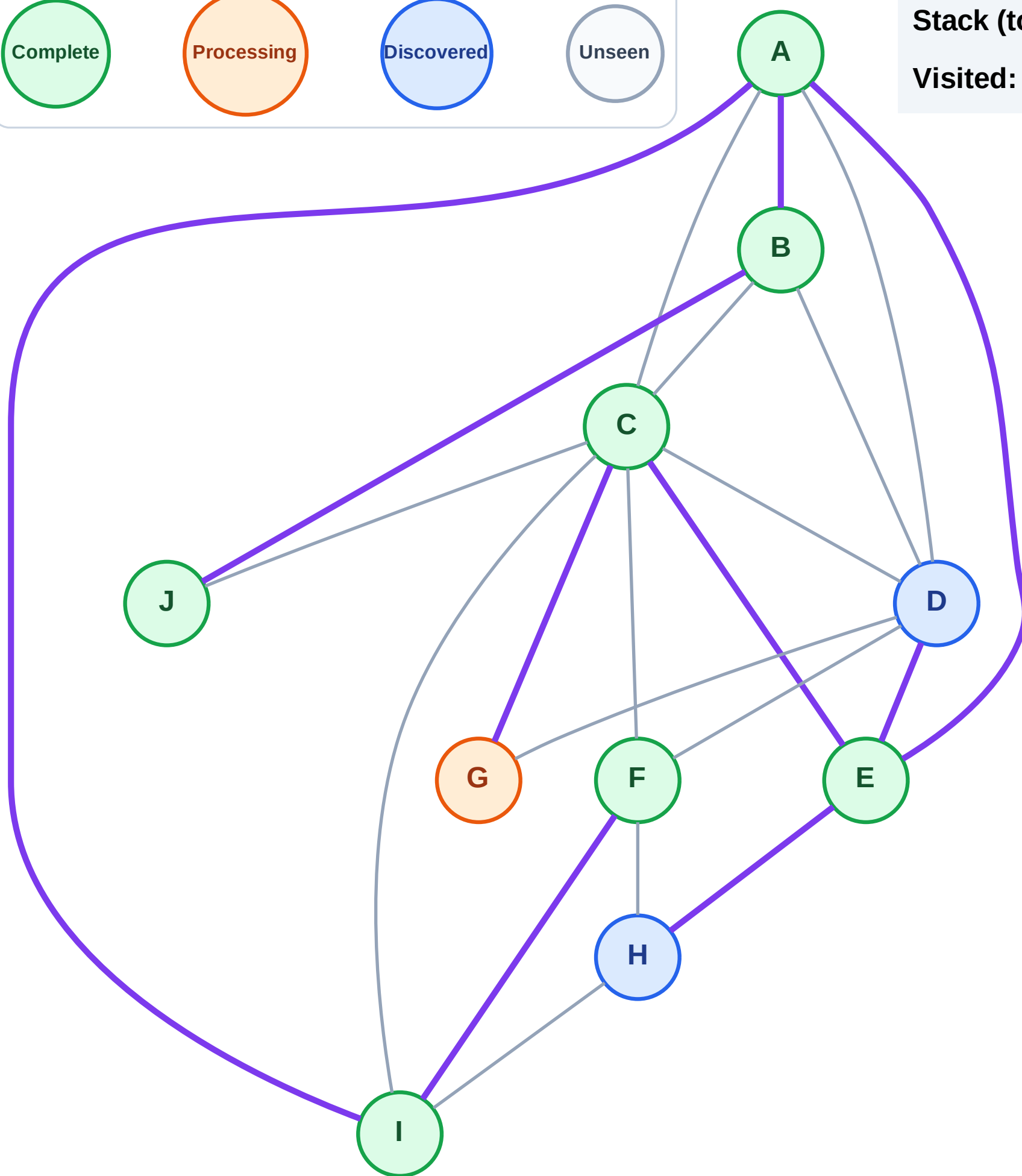
Step 16: Process node G

Legend



Stack (top at right): H | D

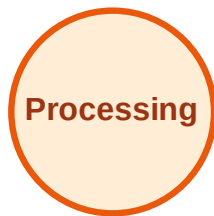
Visited: A, B, C, E, F, I, J



DFS Traversal

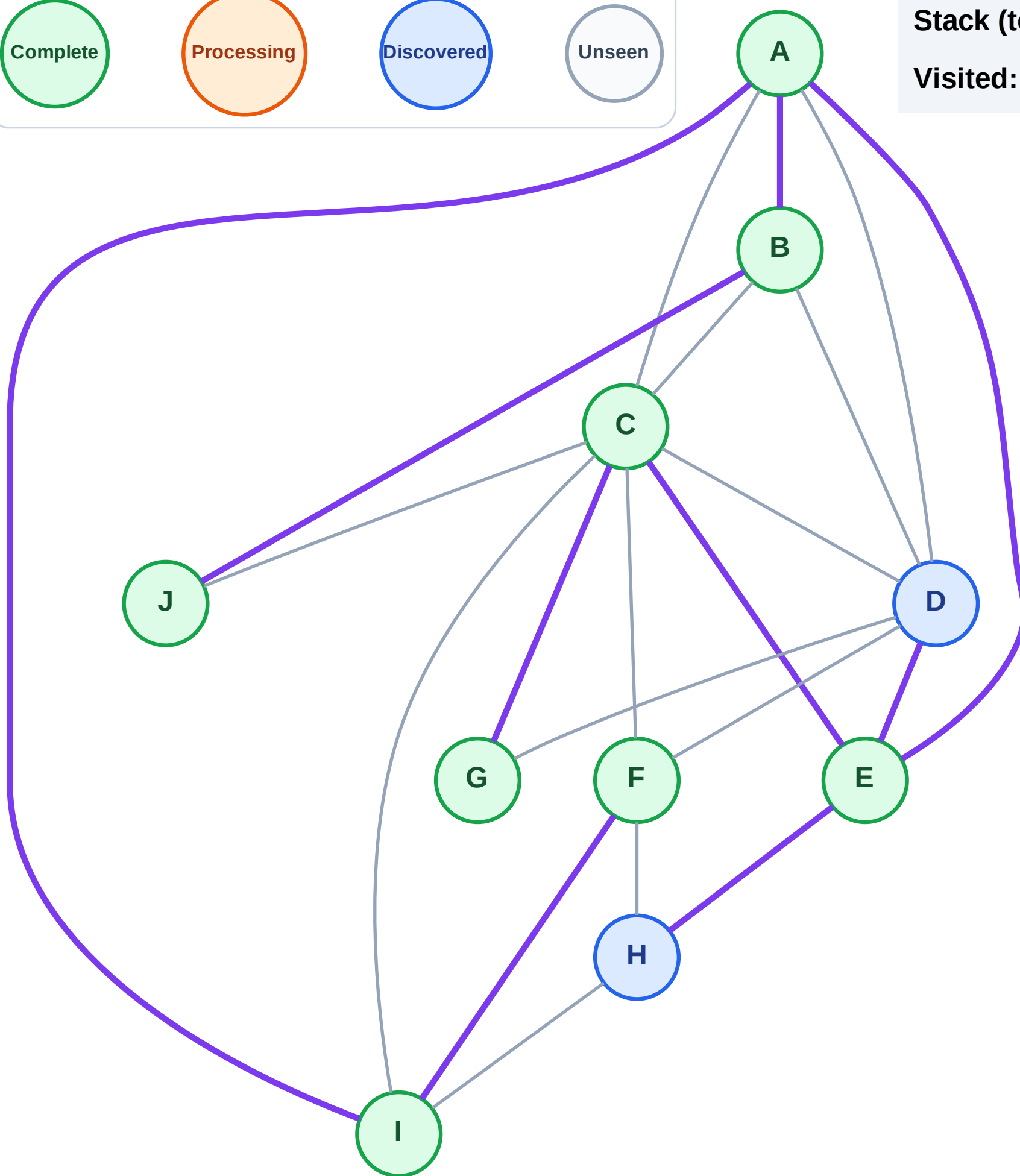
Step 17: No new neighbors from G

Legend



Stack (top at right): H | D

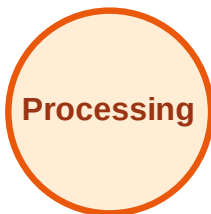
Visited: A, B, C, E, F, G, I, J



DFS Traversal

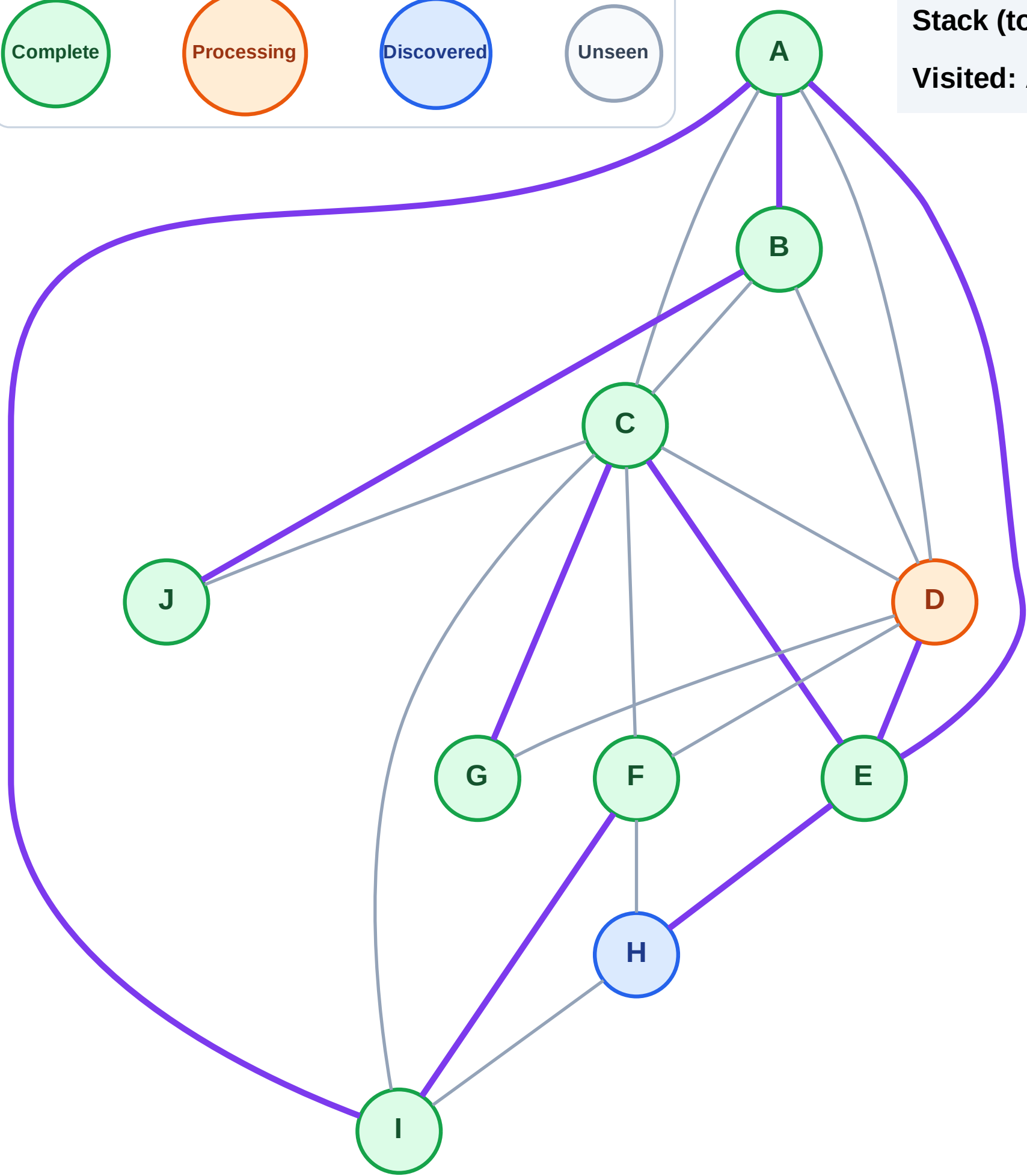
Step 18: Process node D

Legend



Stack (top at right): H

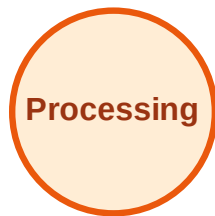
Visited: A, B, C, E, F, G, I, J



DFS Traversal

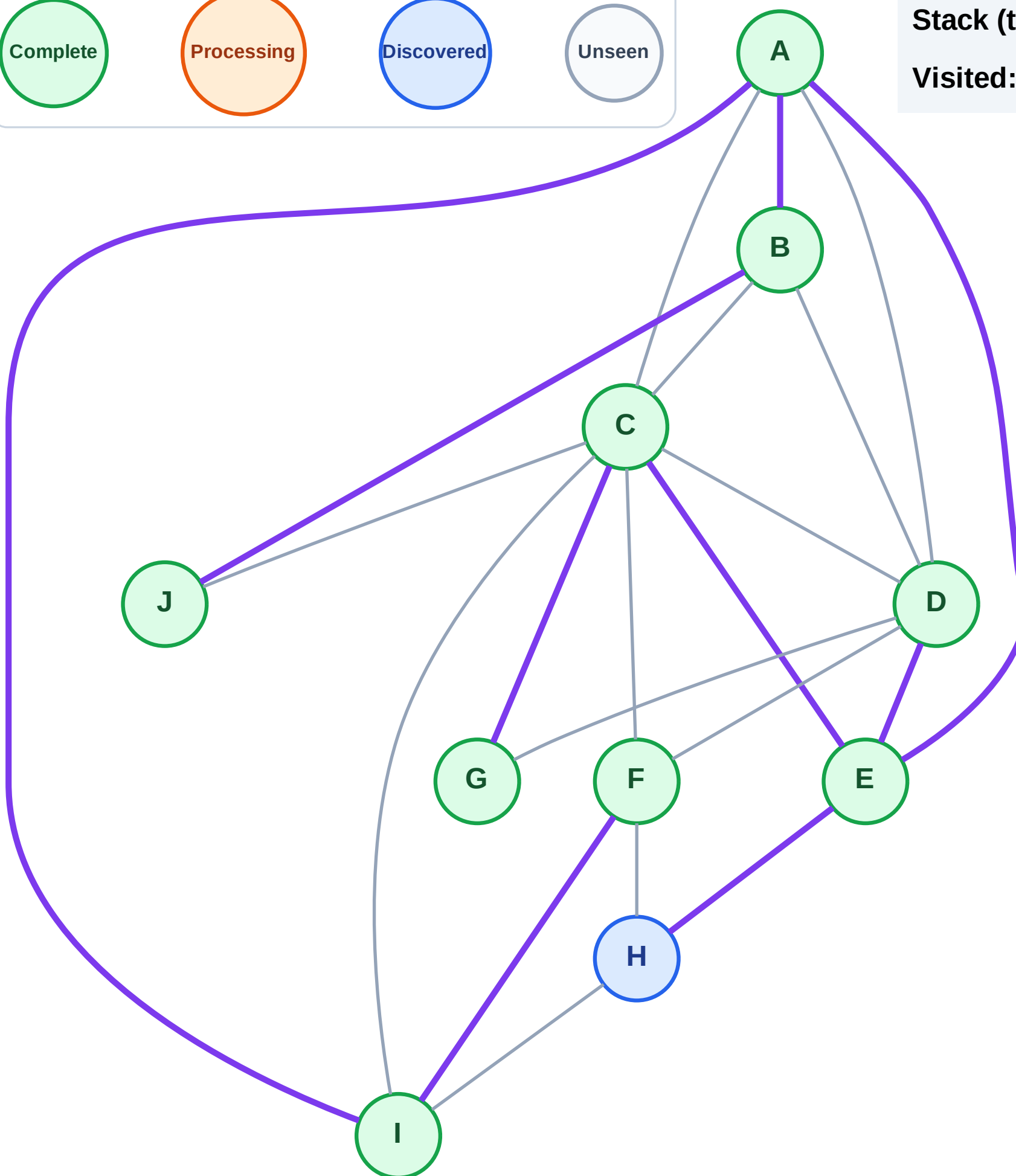
Step 19: No new neighbors from D

Legend



Stack (top at right): H

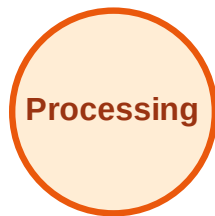
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

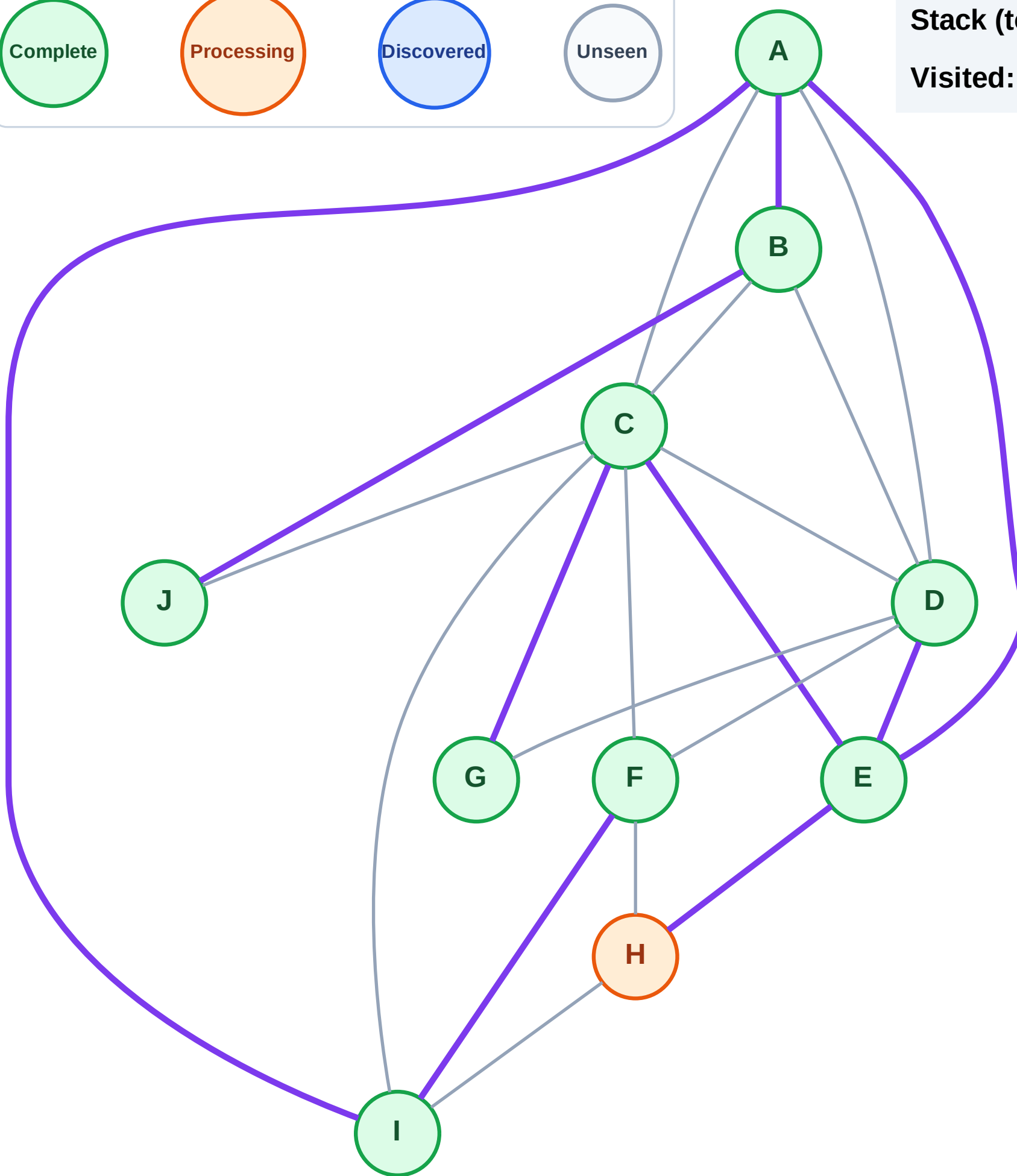
Step 20: Process node H

Legend



Stack (top at right): (empty)

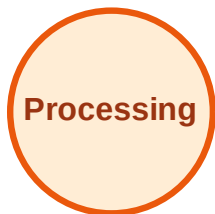
Visited: A, B, C, D, E, F, G, I, J



DFS Traversal

Step 21: No new neighbors from H

Legend



Stack (top at right): (empty)

Visited: A, B, C, D, E, F, G, H, I, J

